



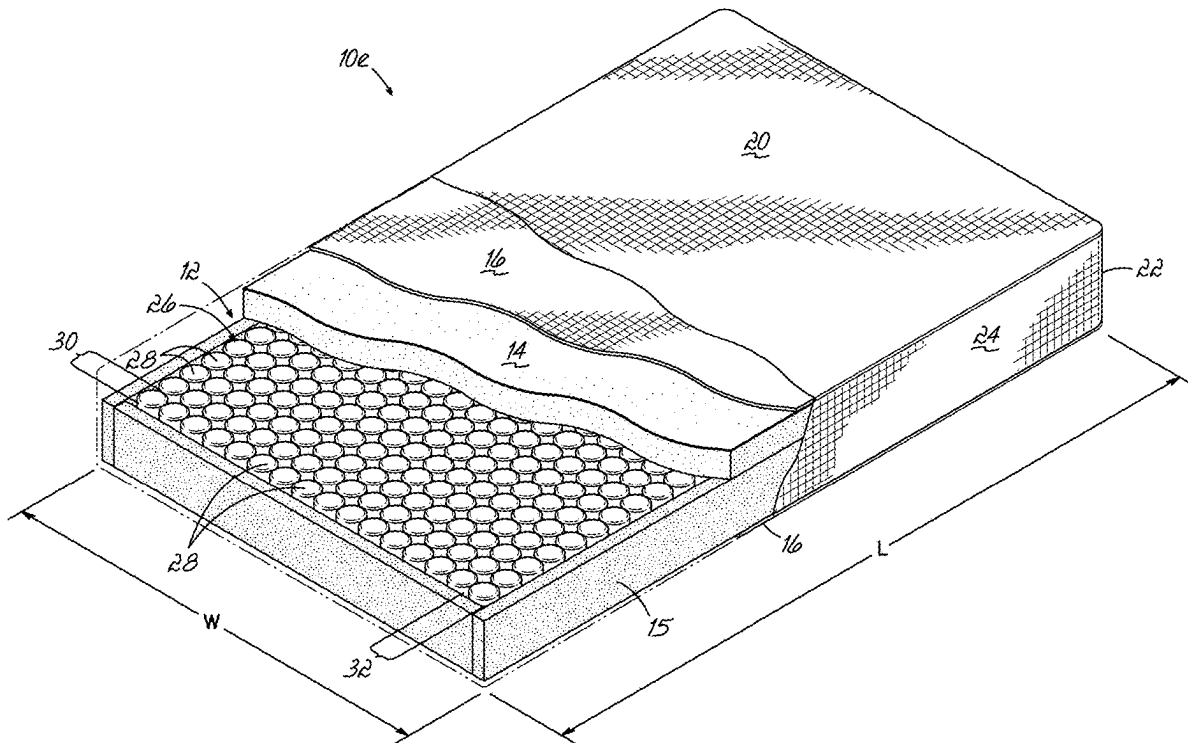
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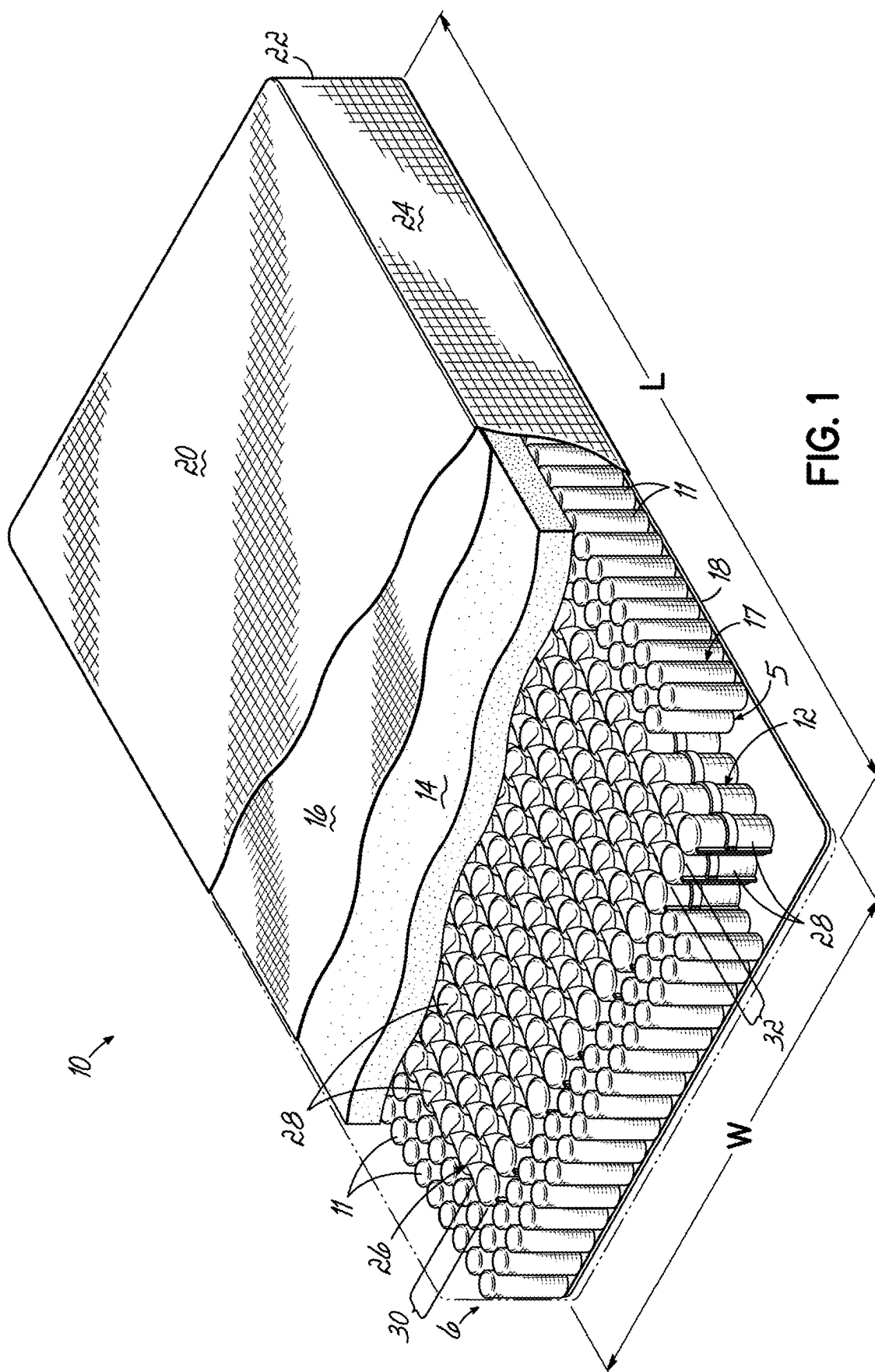
(19) **United States**(12) **Patent Application Publication**
Peterson(10) **Pub. No.: US 2025/0268393 A1**(43) **Pub. Date: Aug. 28, 2025**(54) **POCKETED FOAM FILLED BUCKLING
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Company, South Gate, CA (US)**(21) Appl. No.: **18/808,170**(22) Filed: **Aug. 19, 2024****Related U.S. Application Data**(63) Continuation-in-part of application No. 18/586,716,
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(2013.01)

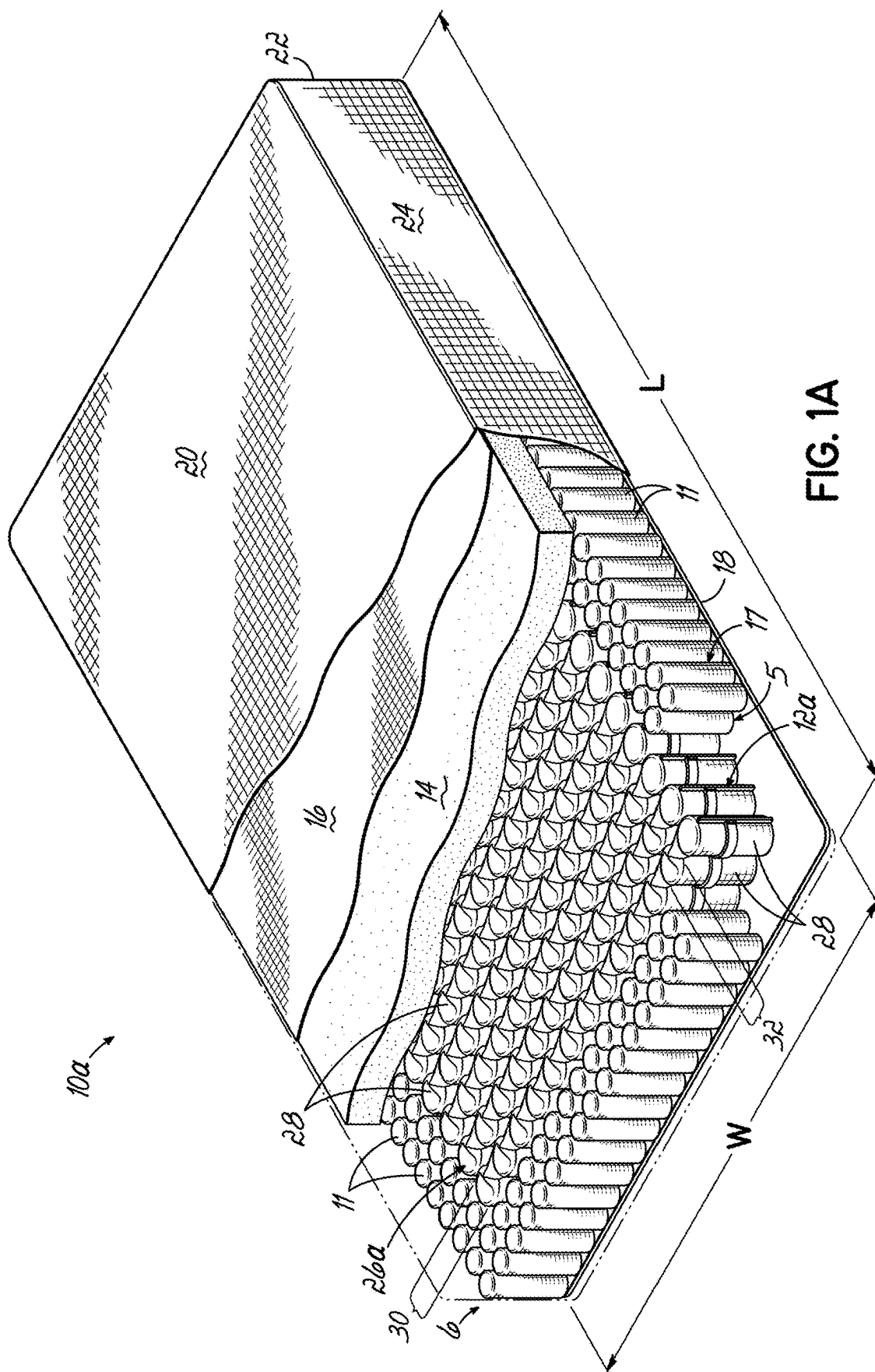
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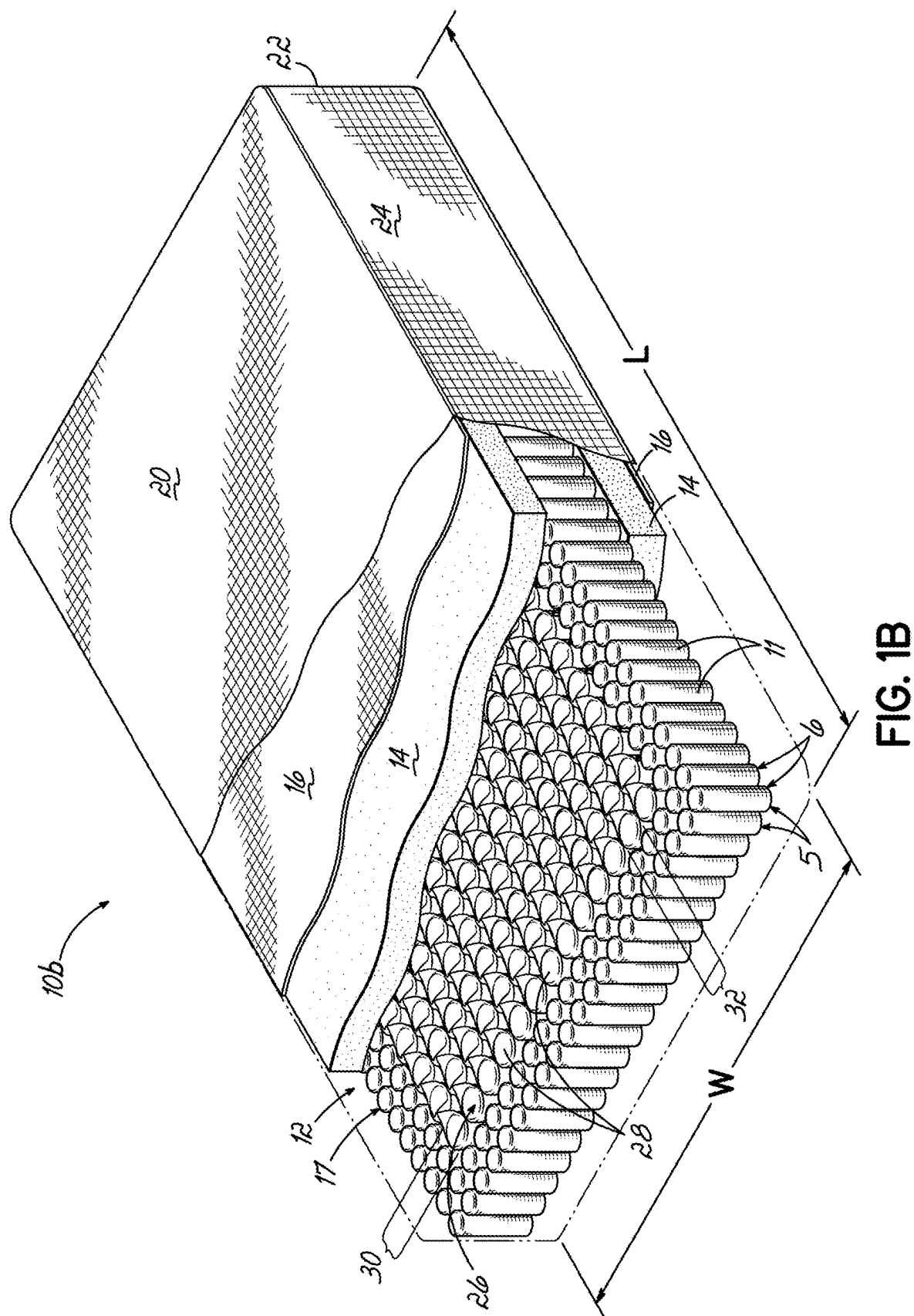
ABSTRACT

A pocketed buckling member assembly comprises a plurality of parallel strings of pocketed buckling members. Each string is joined to at least one adjacent string. Each string comprising one piece of fabric folded around the buckling members into first and second opposed plies. Pockets are formed along each string by transverse seams joining the first and second plies. At least one buckling member is positioned in each pocket. At least some of the buckling members may have foam therein.









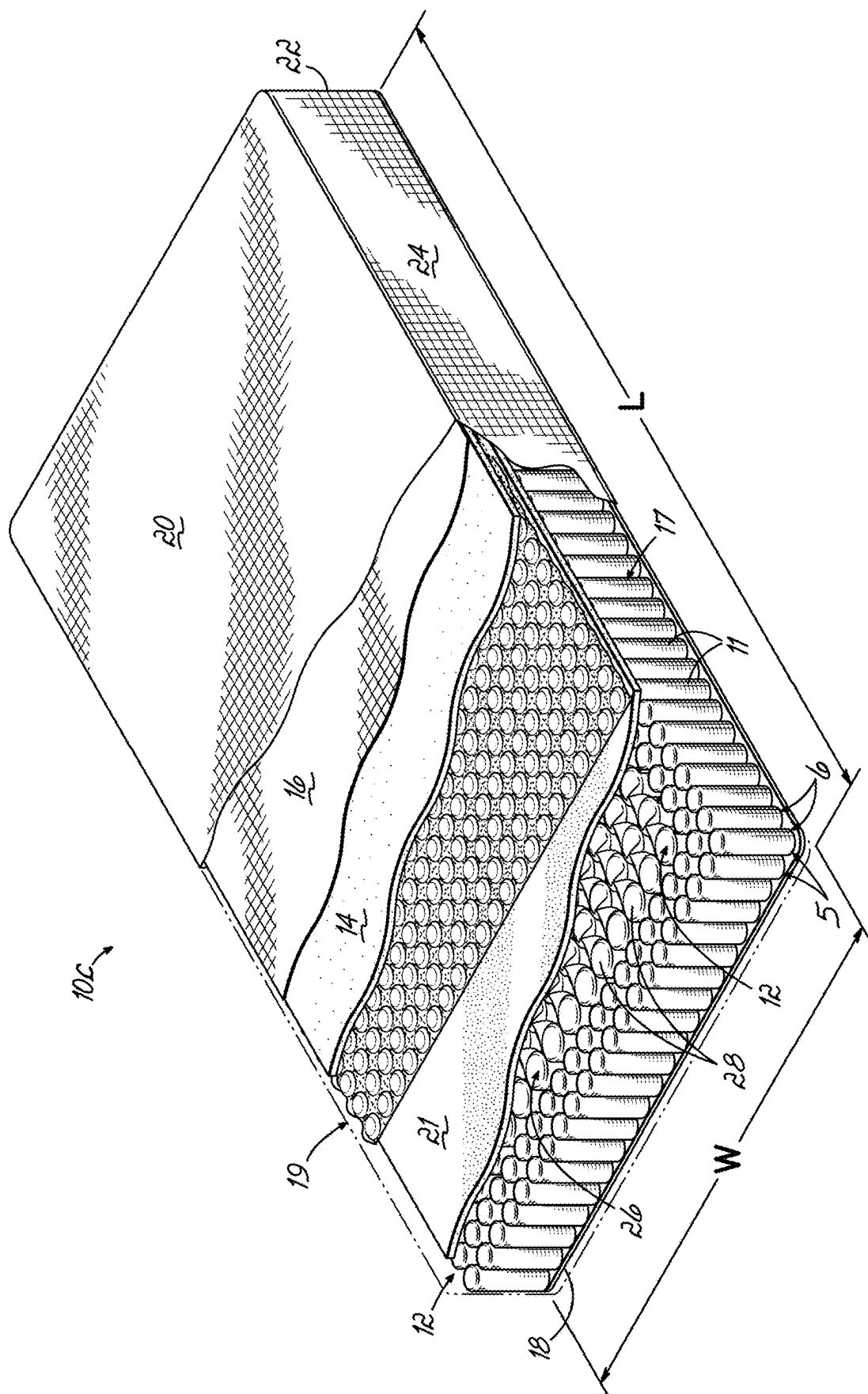
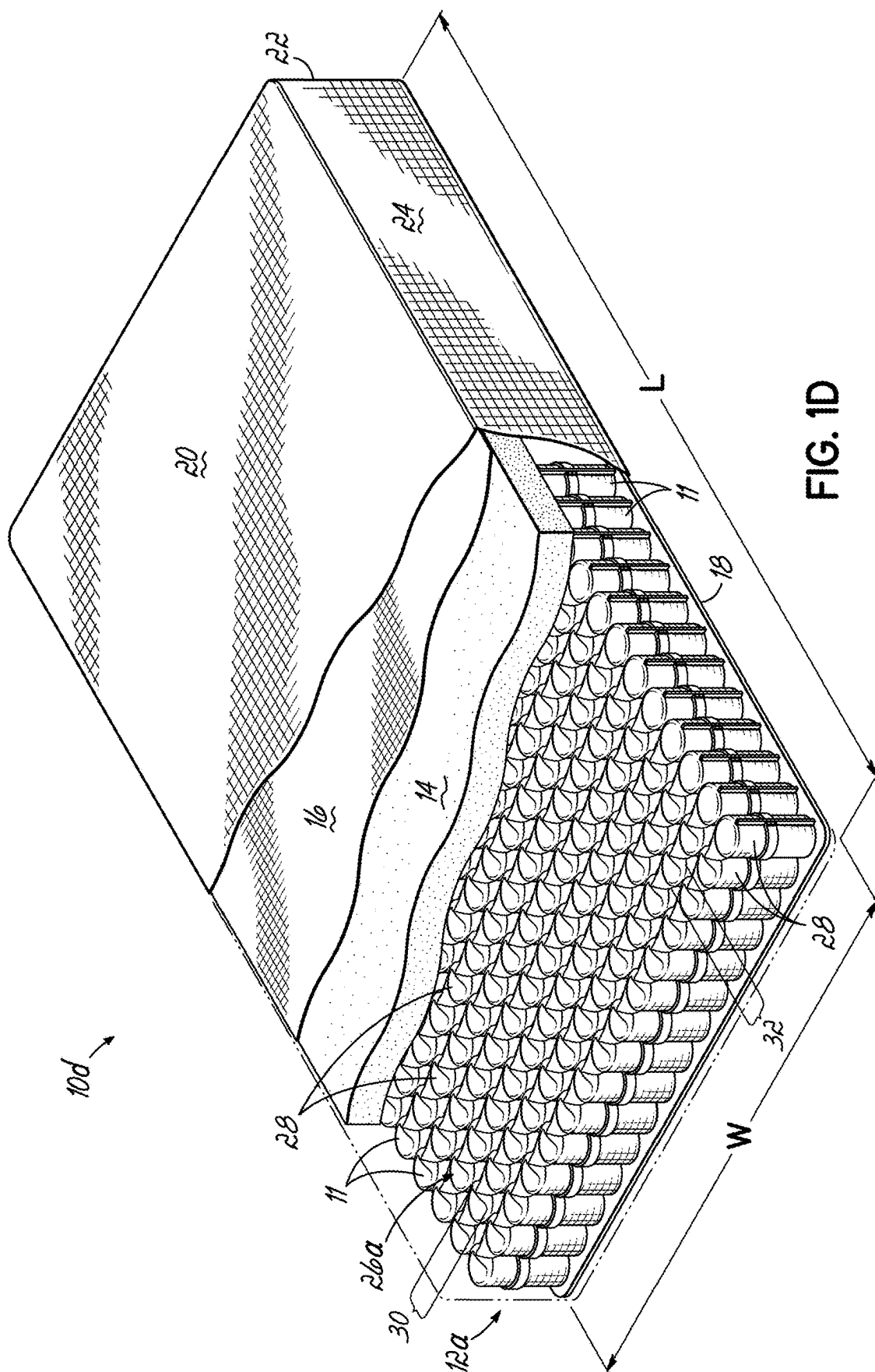


FIG. 1C



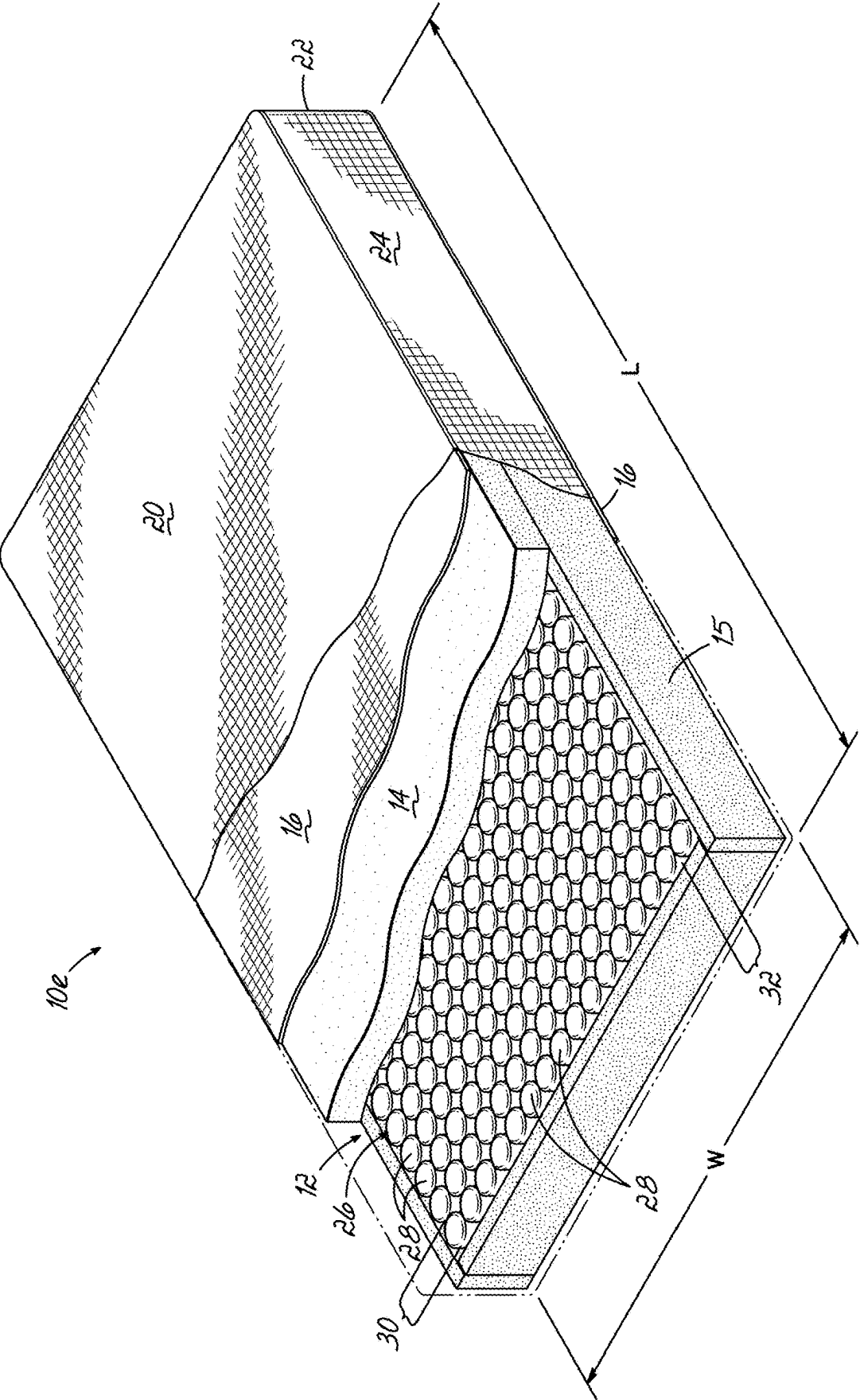
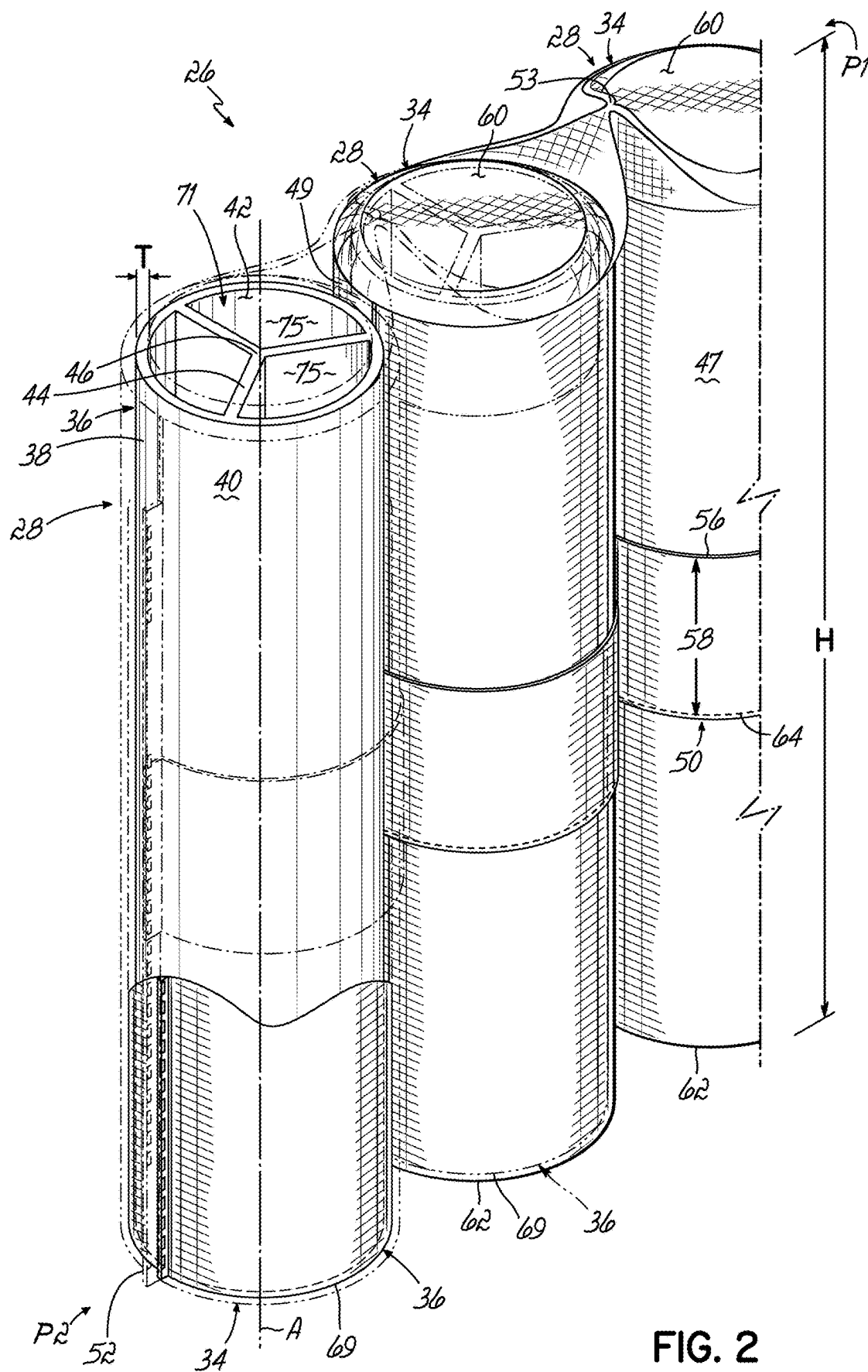


FIG. 1E



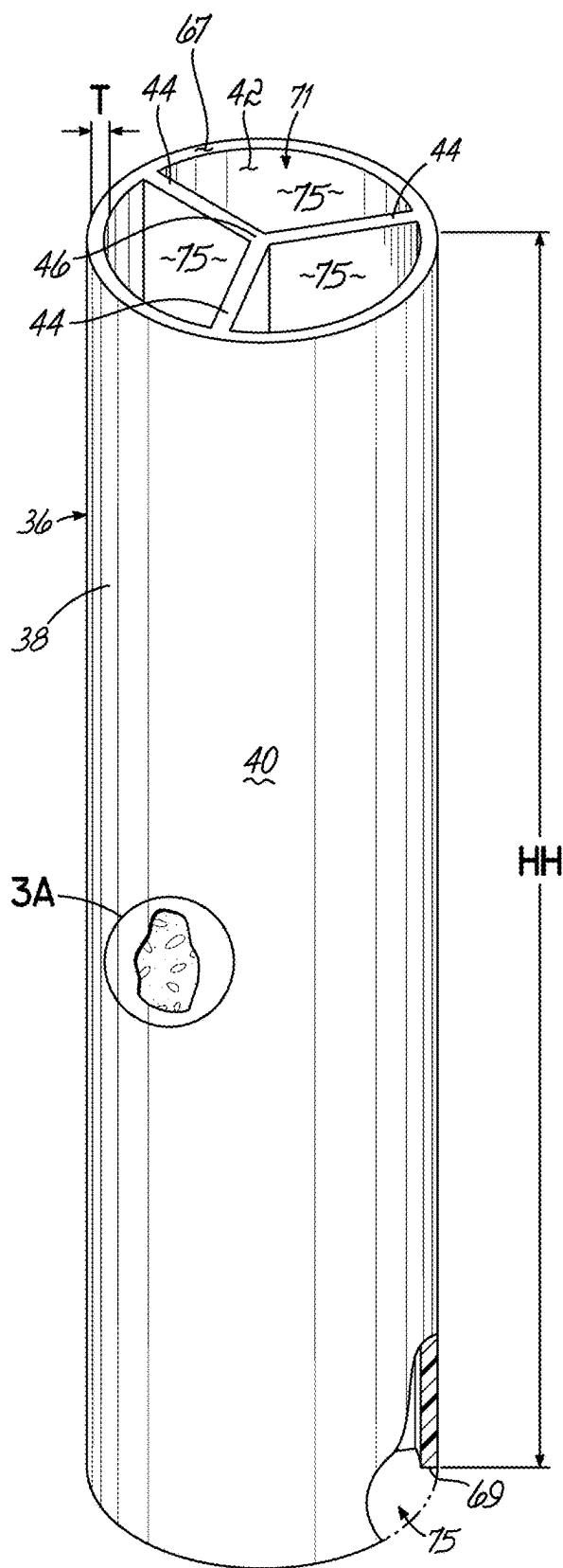


FIG. 3

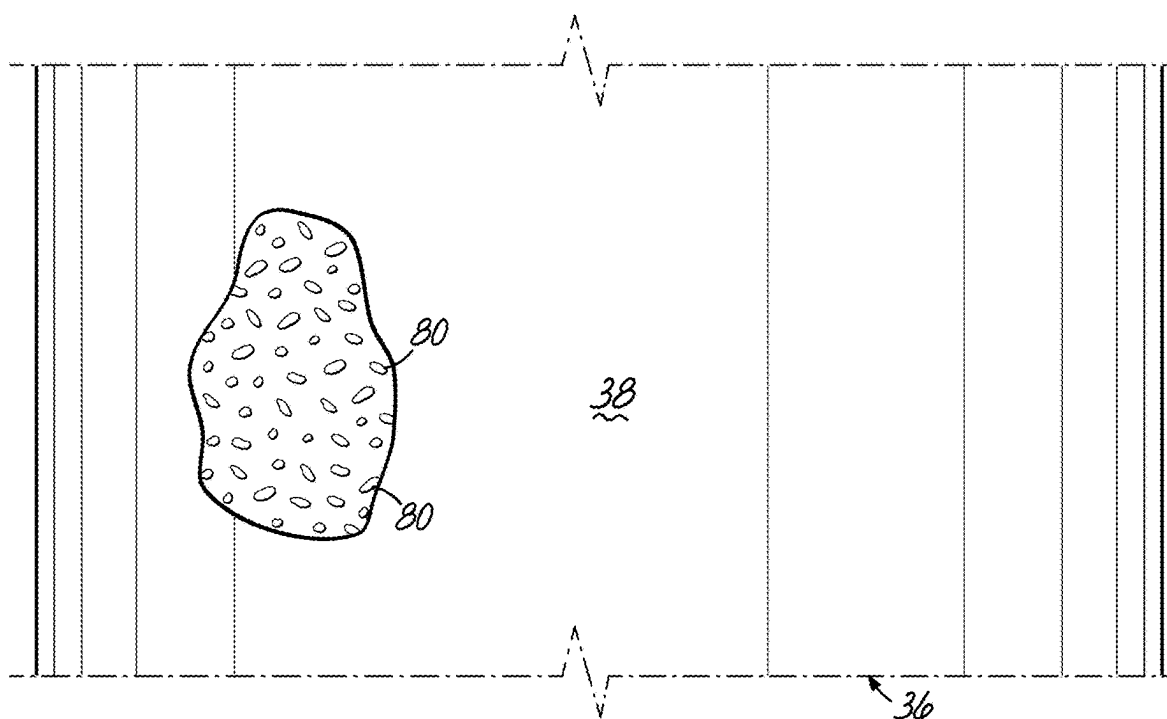


FIG. 3A

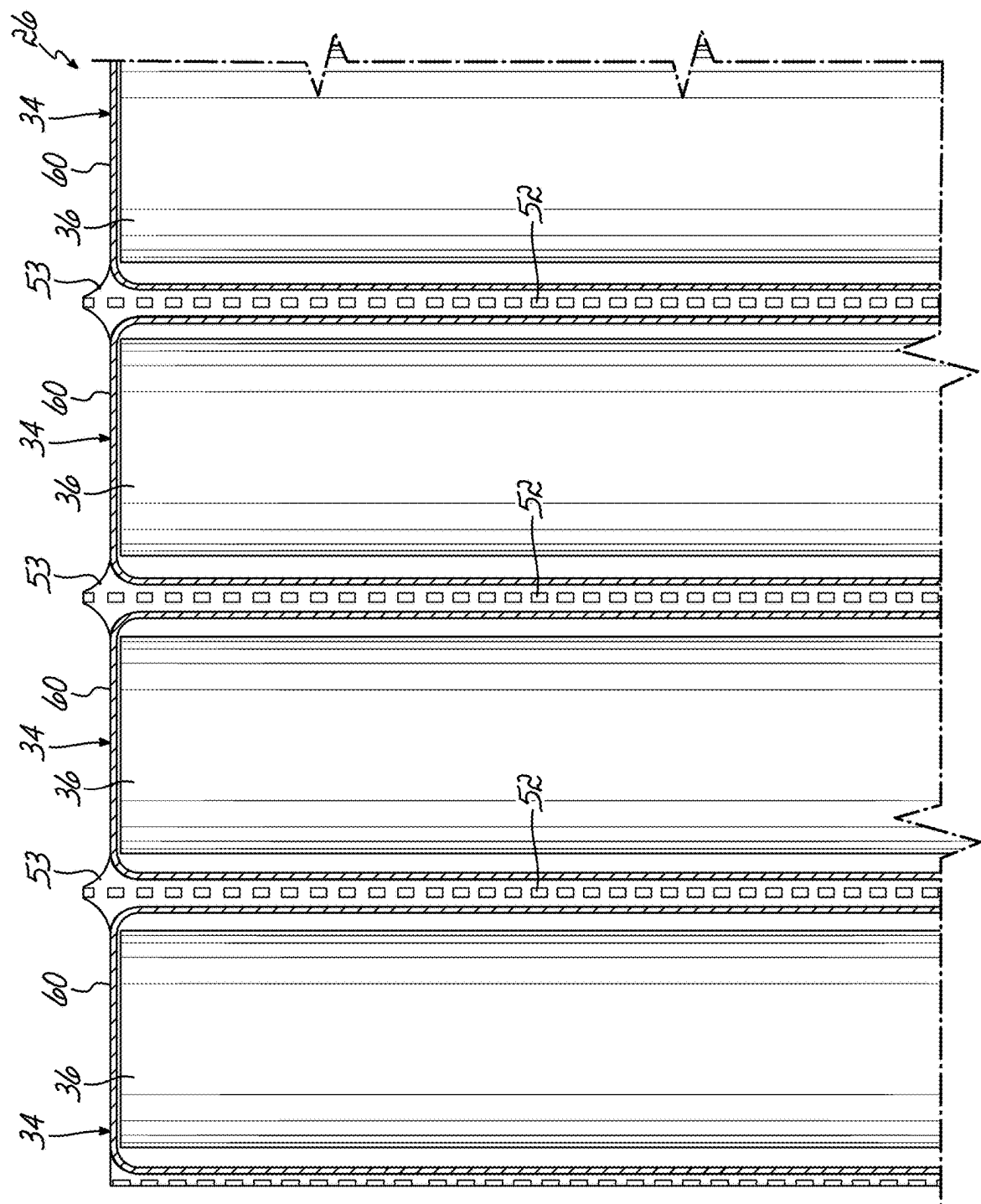
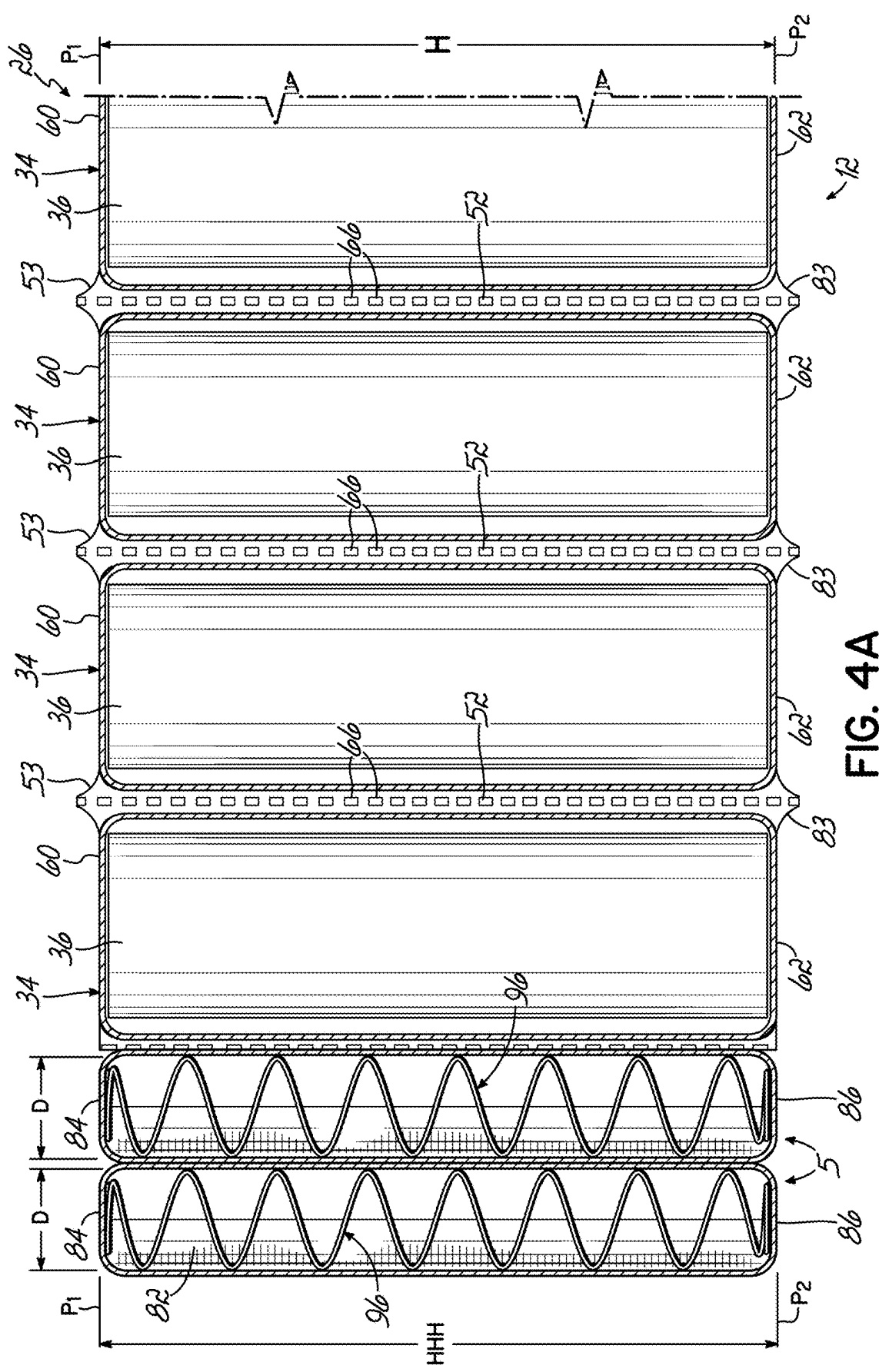
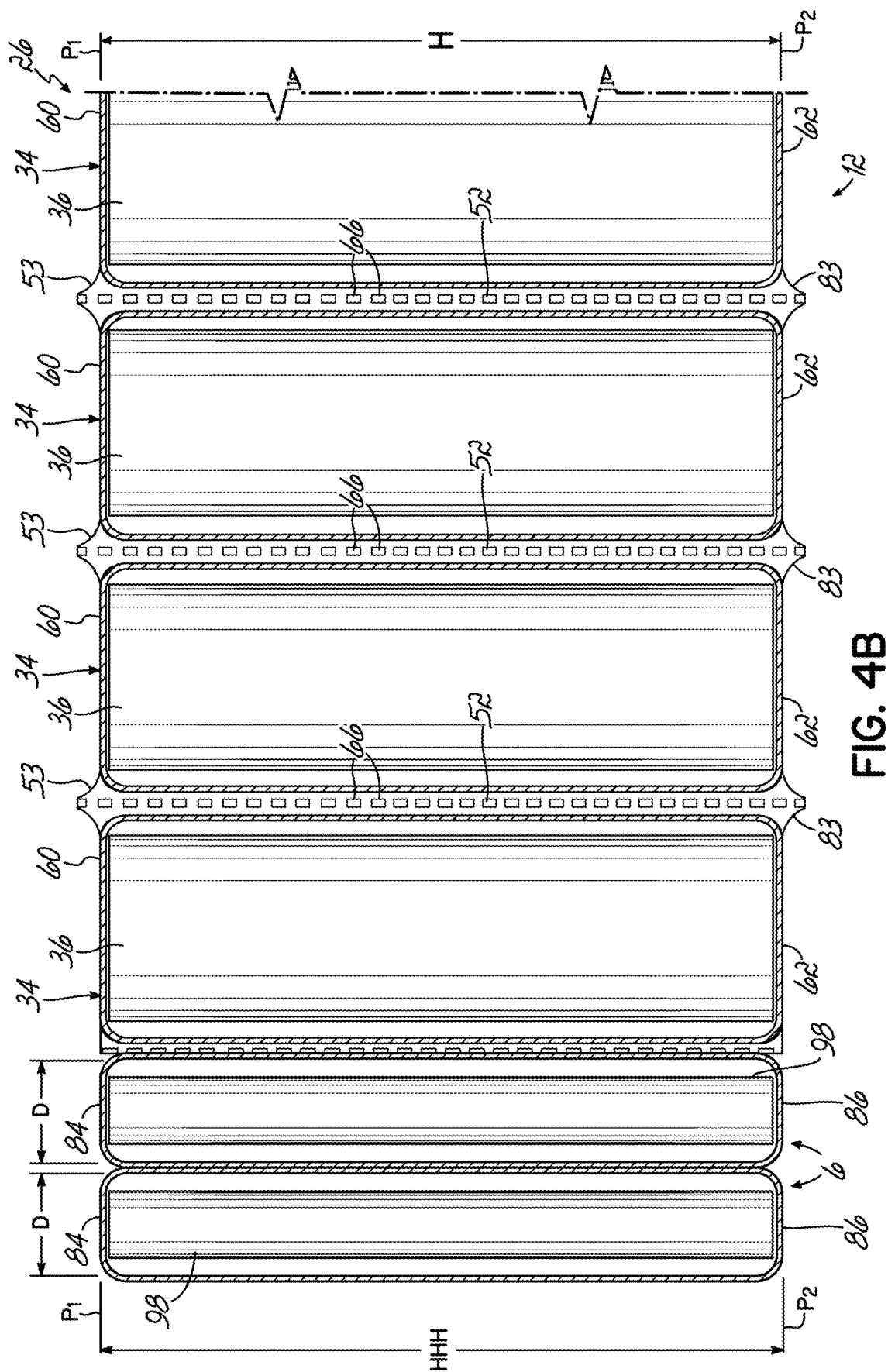


FIG. 4





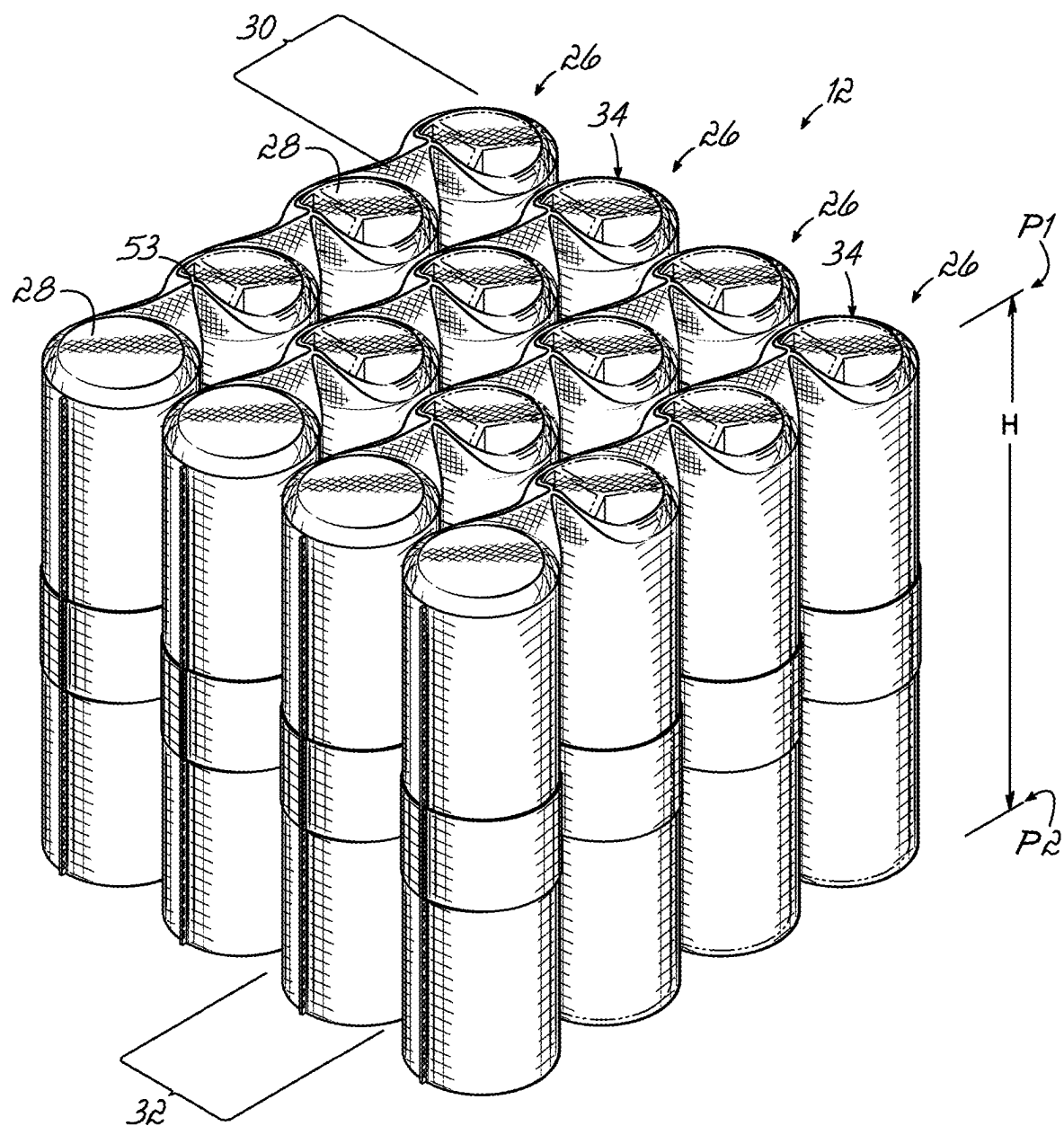


FIG. 5

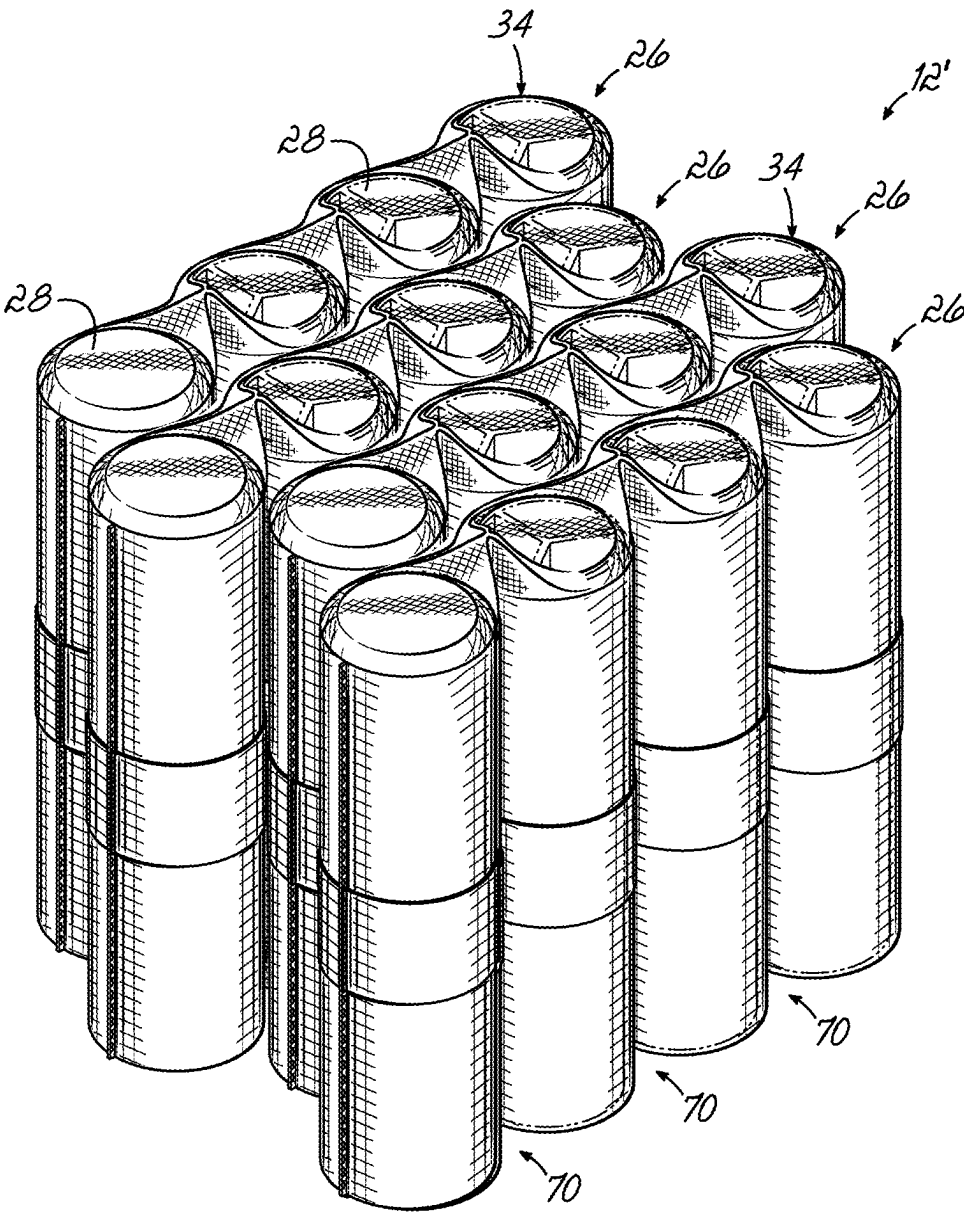


FIG. 6

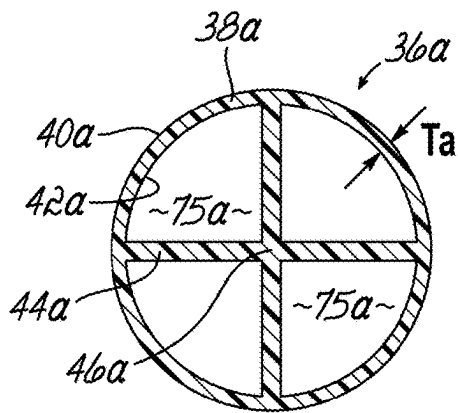


FIG. 7A

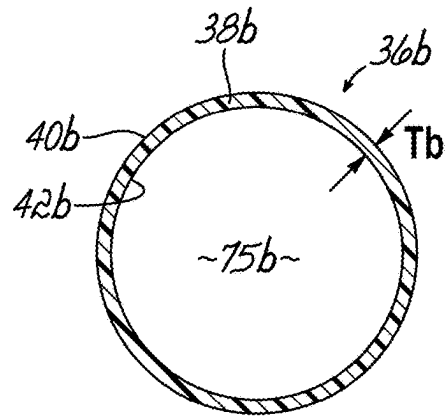


FIG. 7B

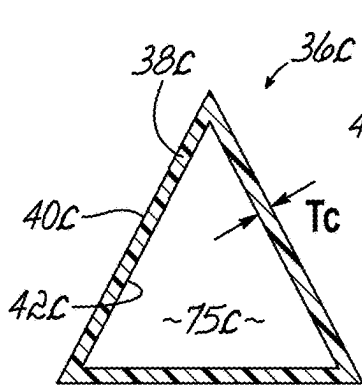


FIG. 7C

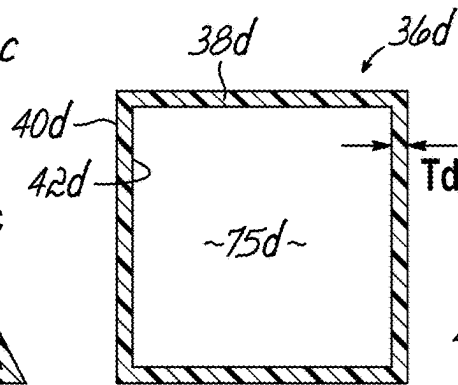


FIG. 7D

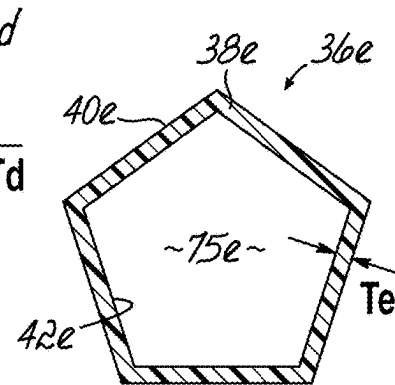


FIG. 7E

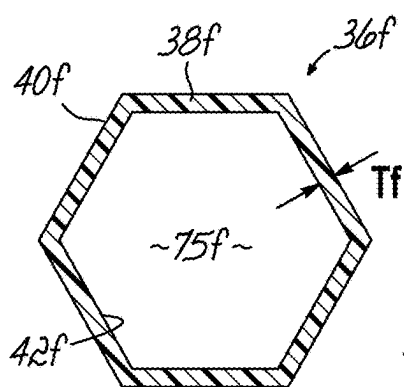


FIG. 7F

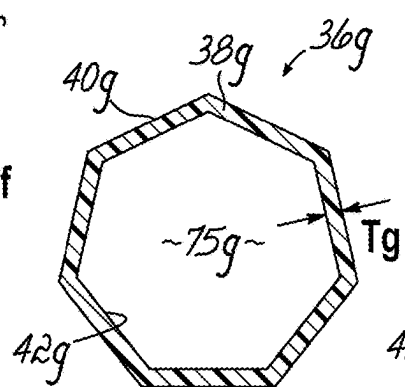


FIG. 7G

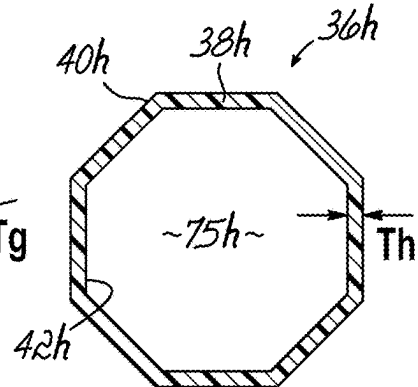


FIG. 7H

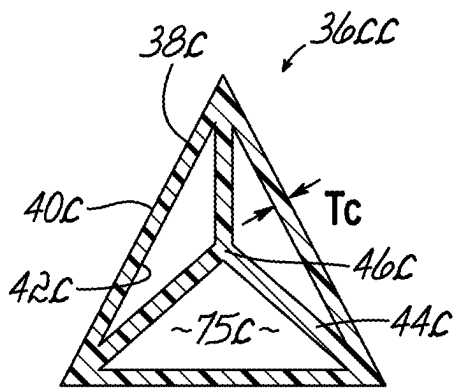


FIG. 7I

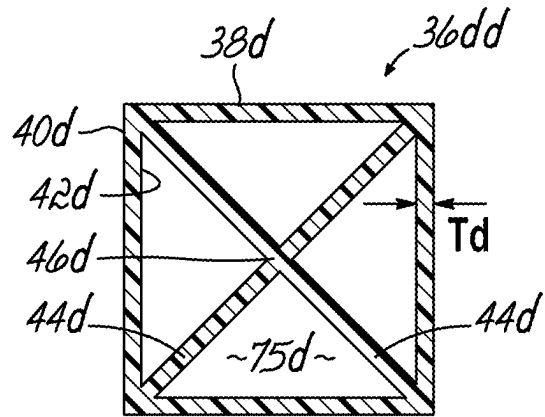


FIG. 7J

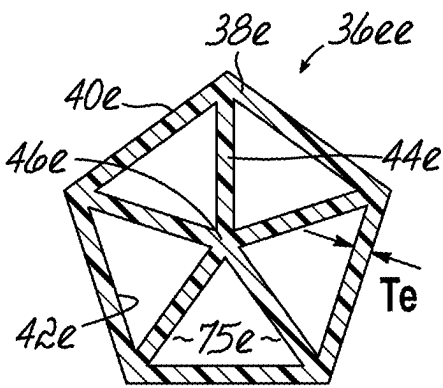


FIG. 7K

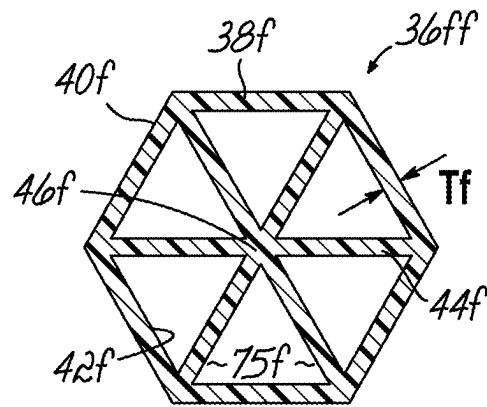


FIG. 7L

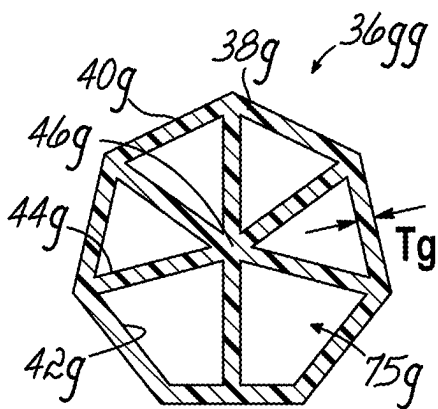


FIG. 7M

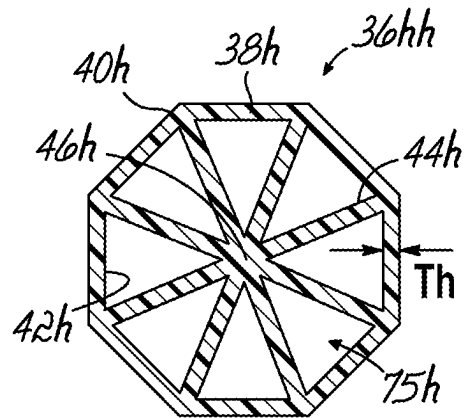


FIG. 7N

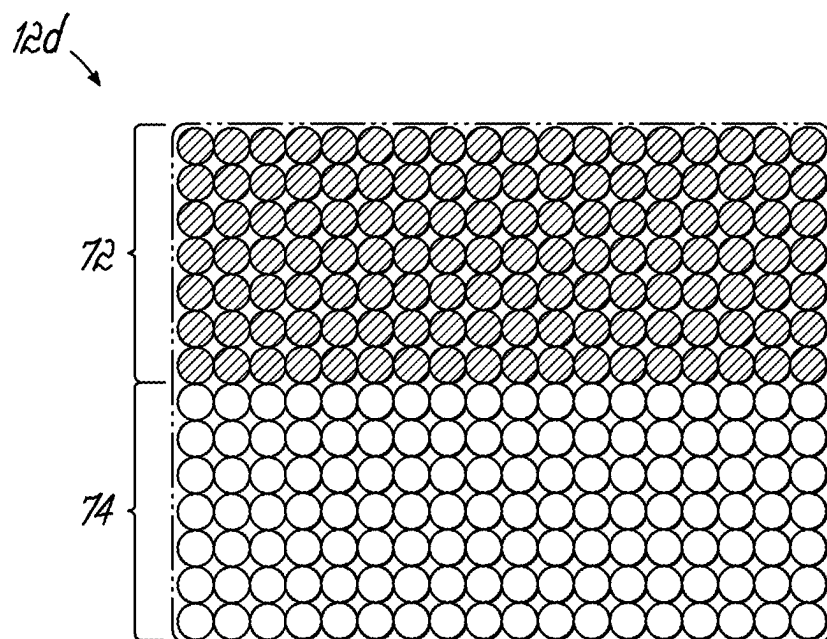


FIG. 8

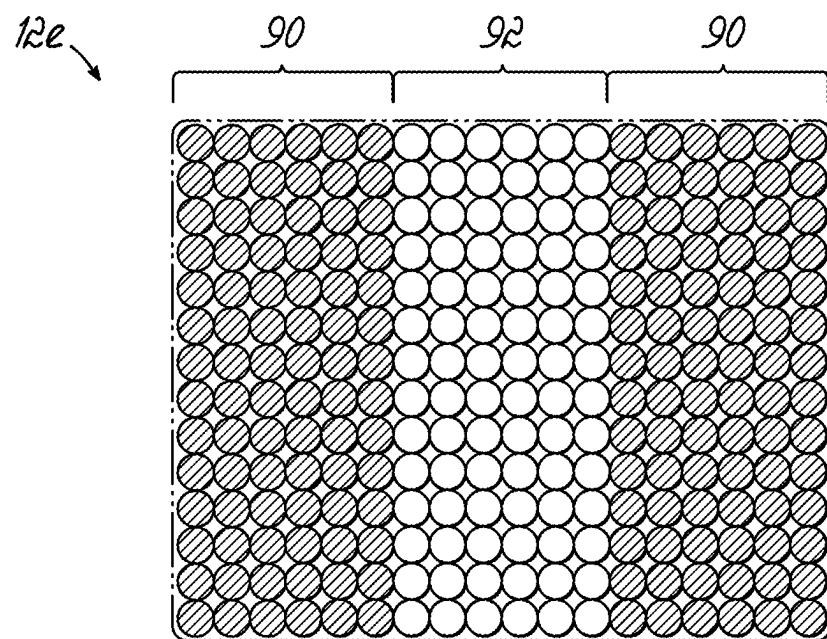


FIG. 9

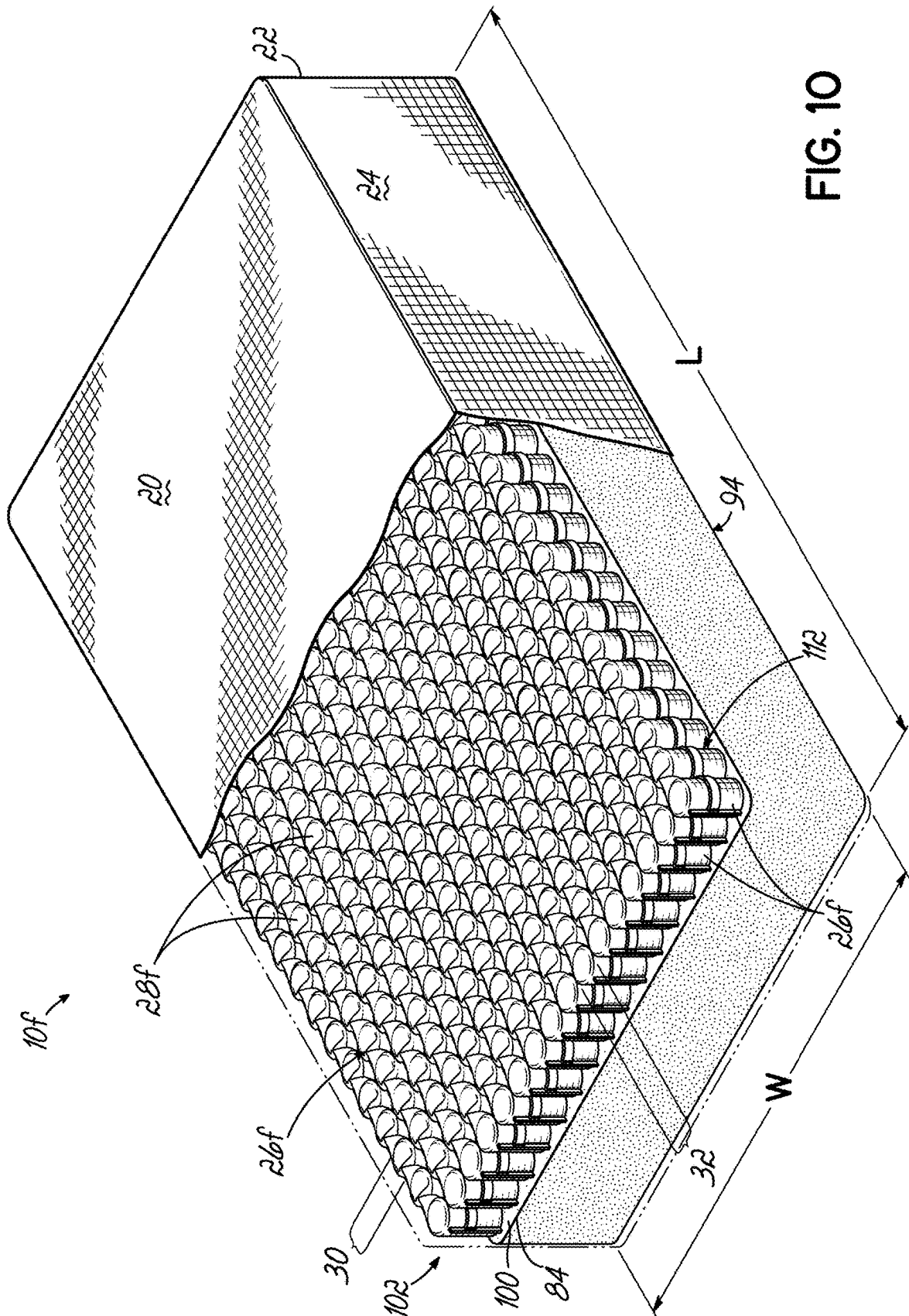


FIG. 10

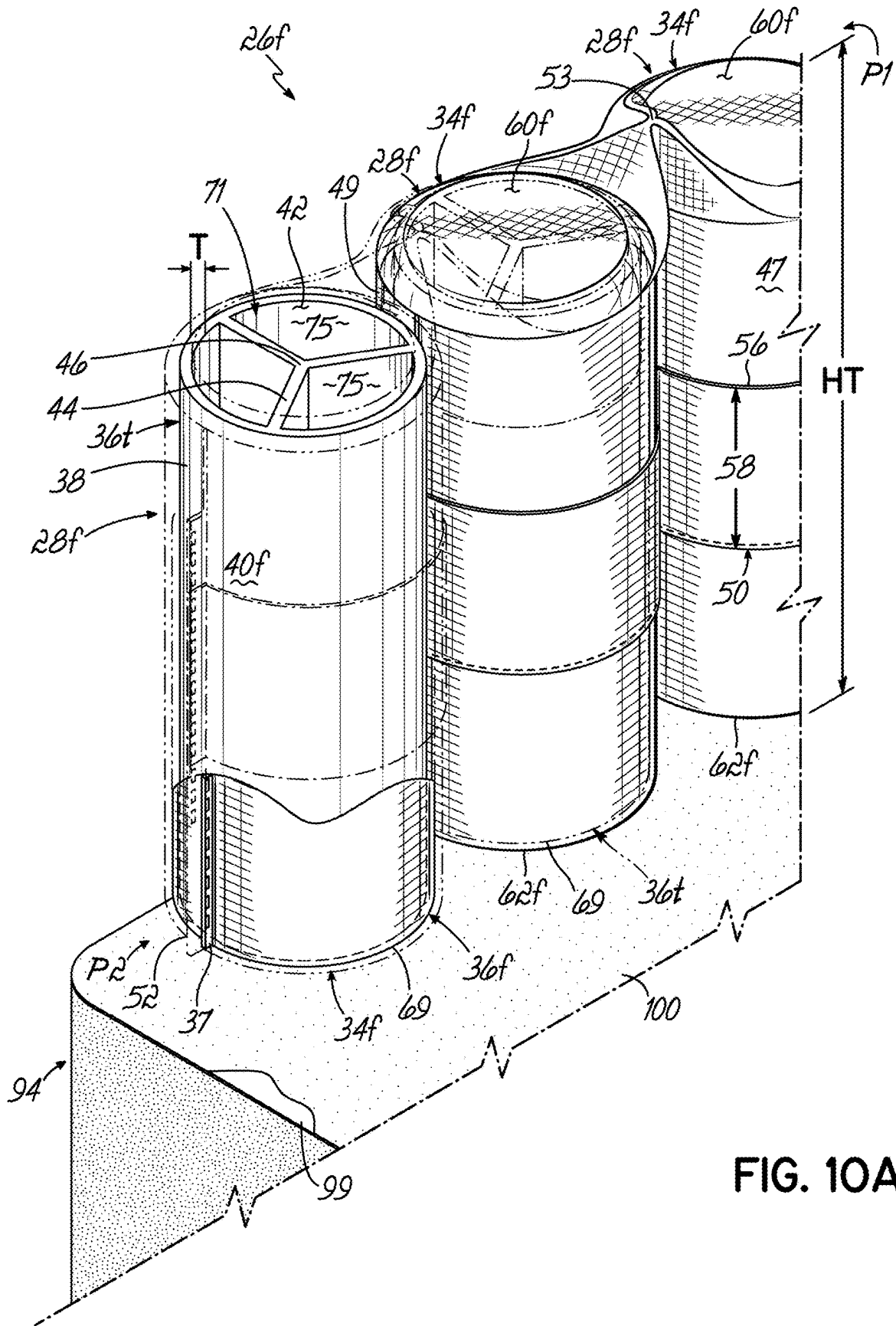


FIG. 10A

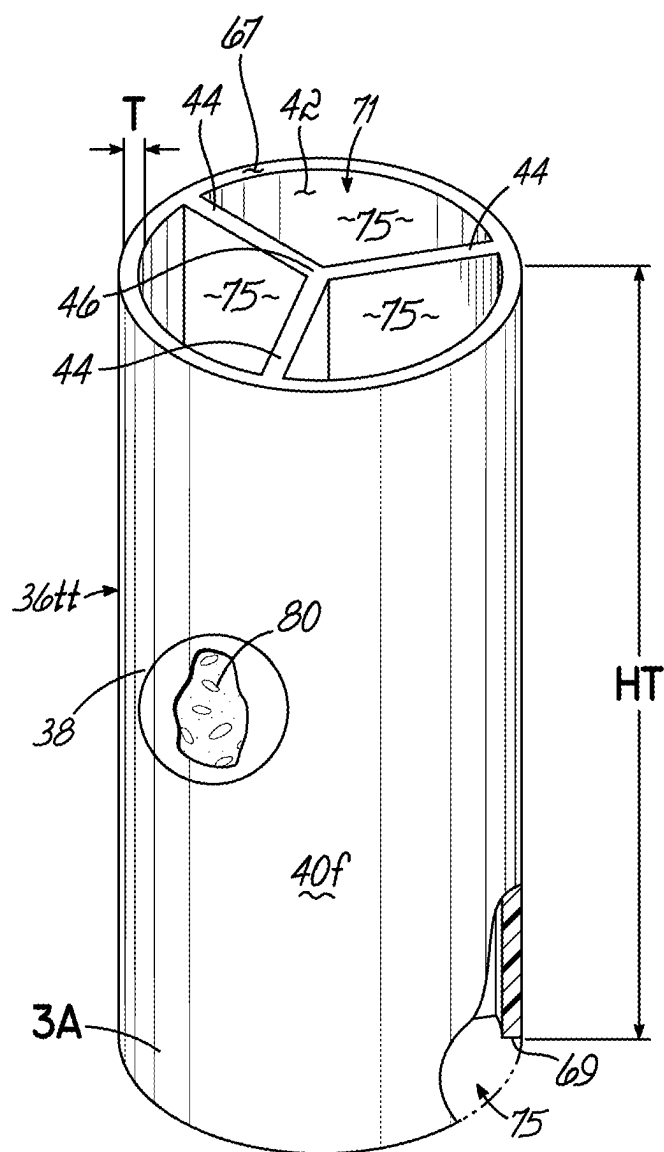


FIG. 10B

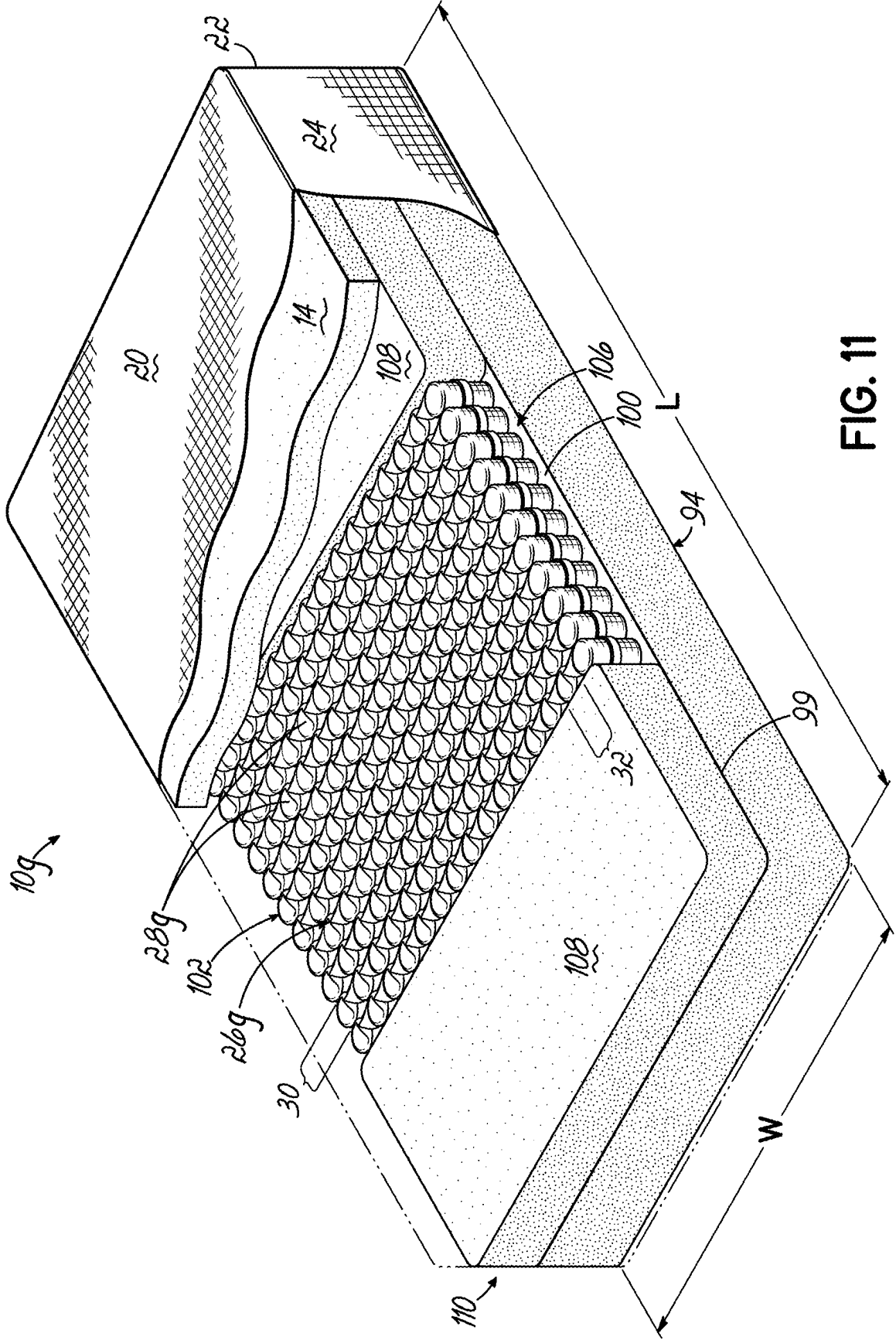


FIG. 11

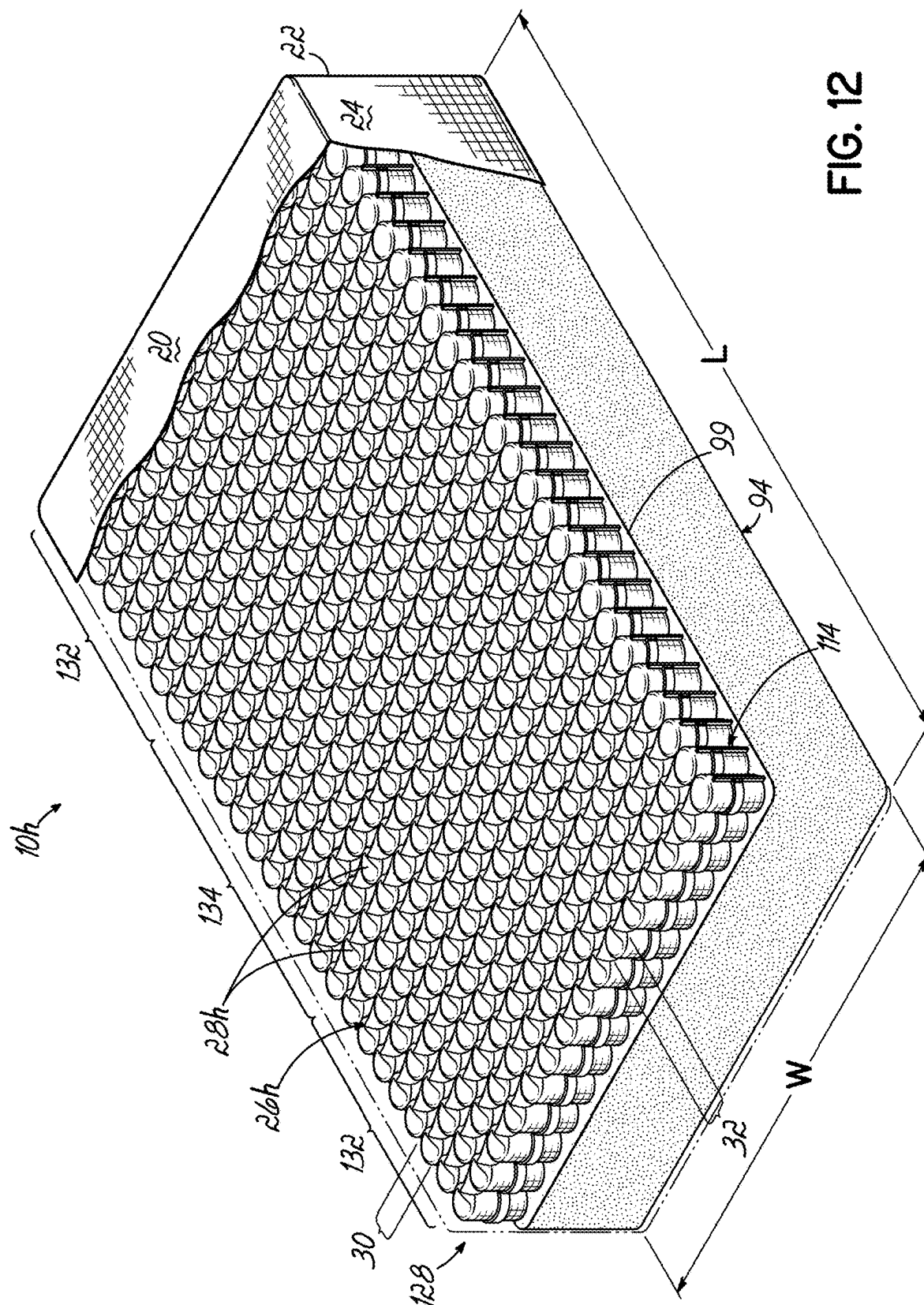


FIG. 12

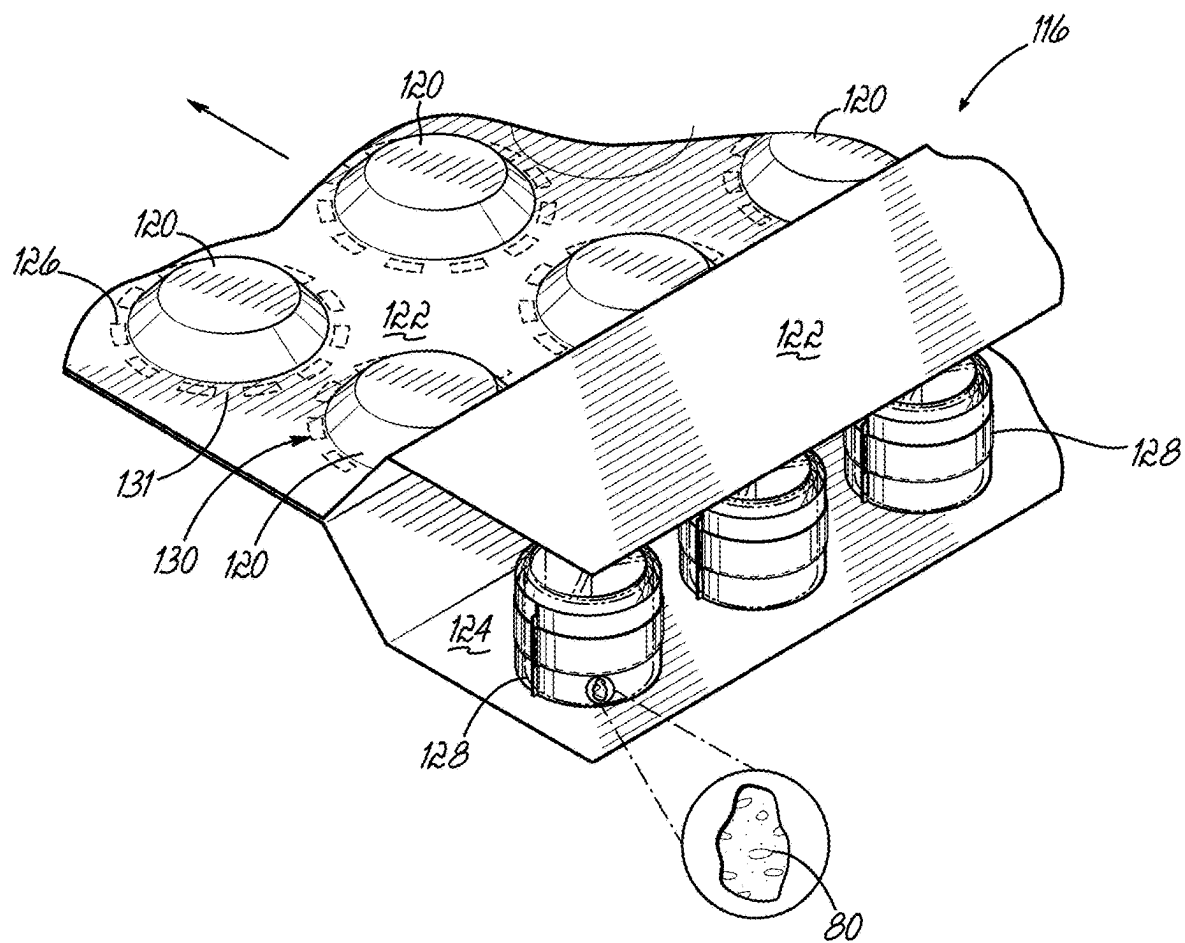


FIG. 13

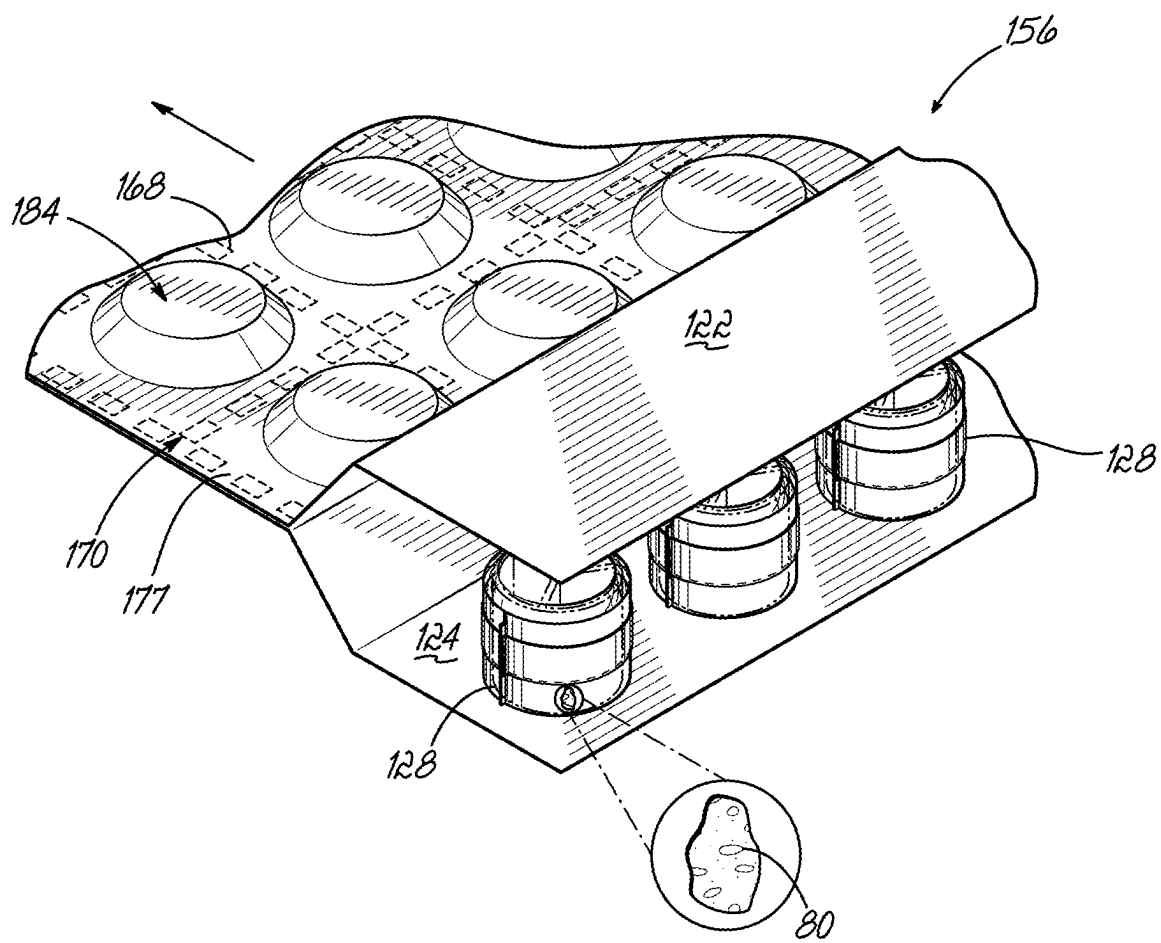


FIG. 14

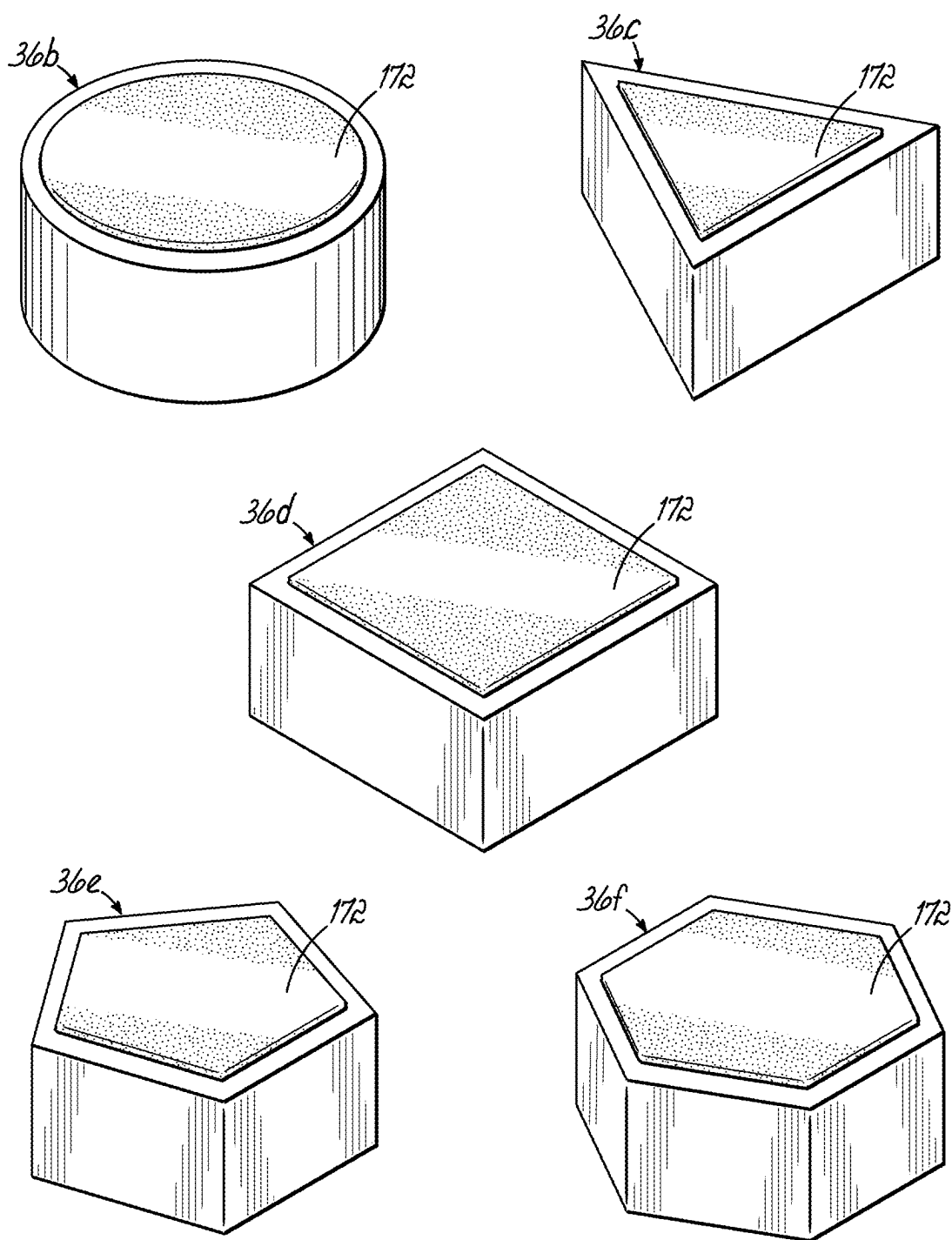


FIG. 15

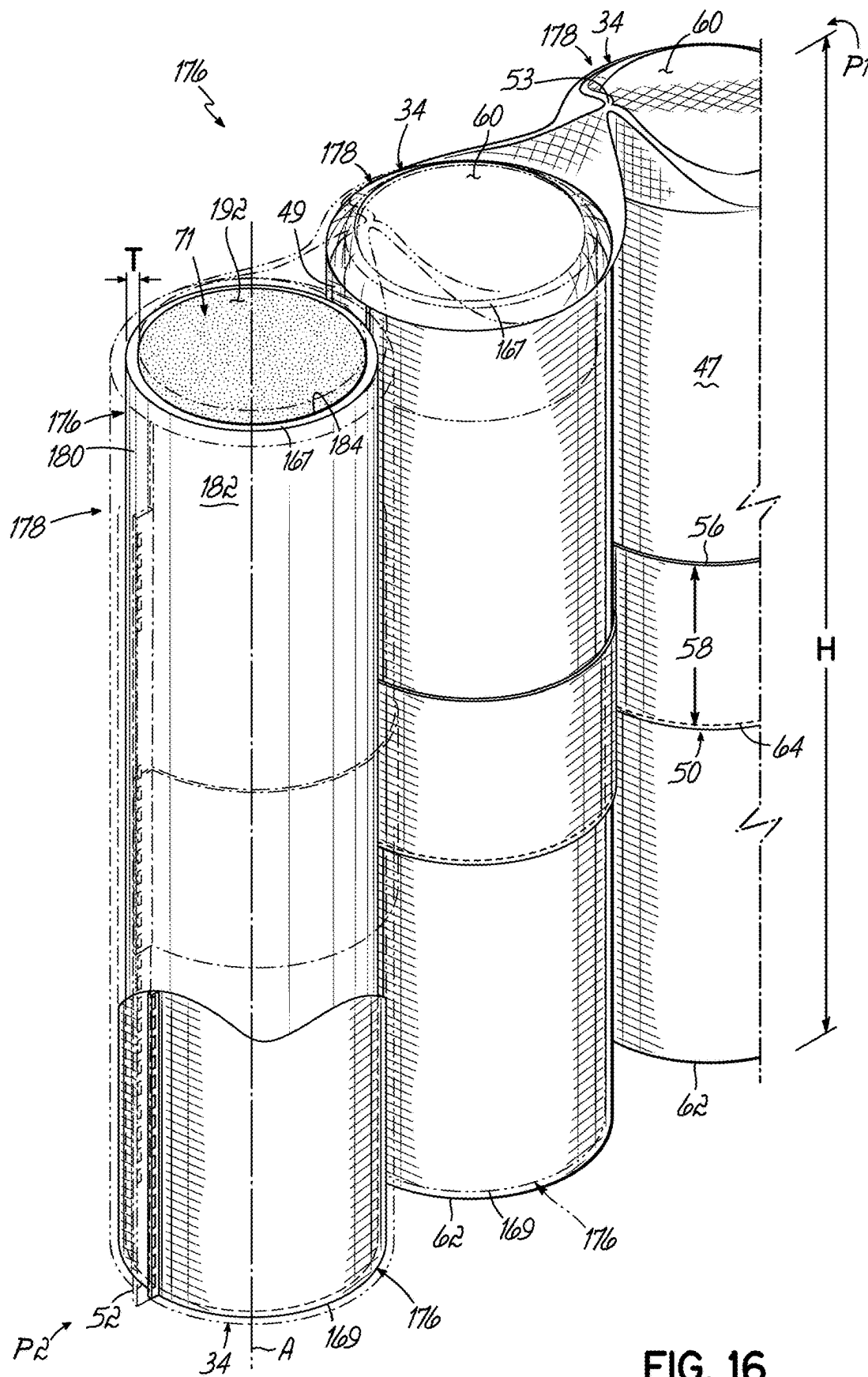


FIG. 16

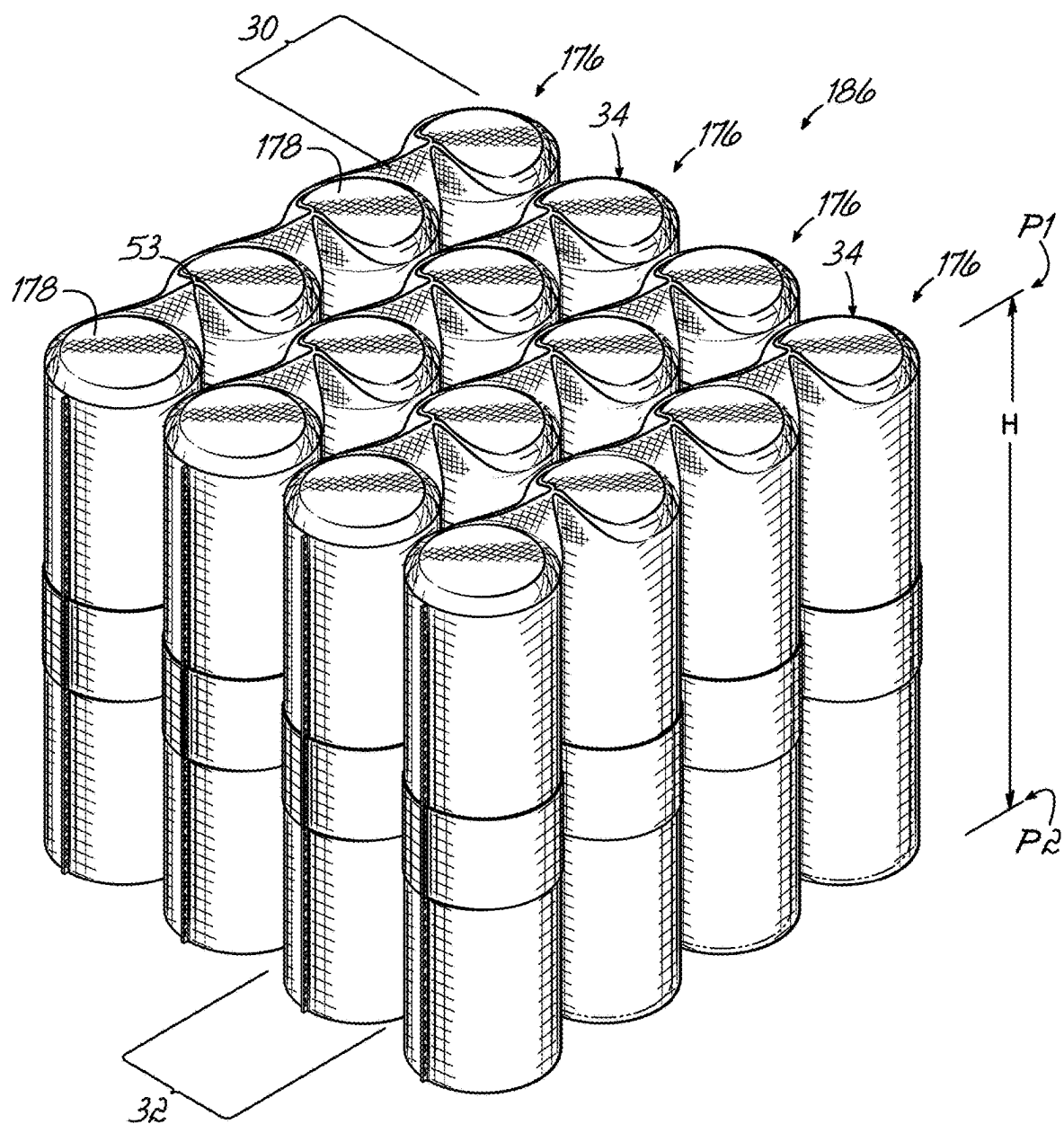
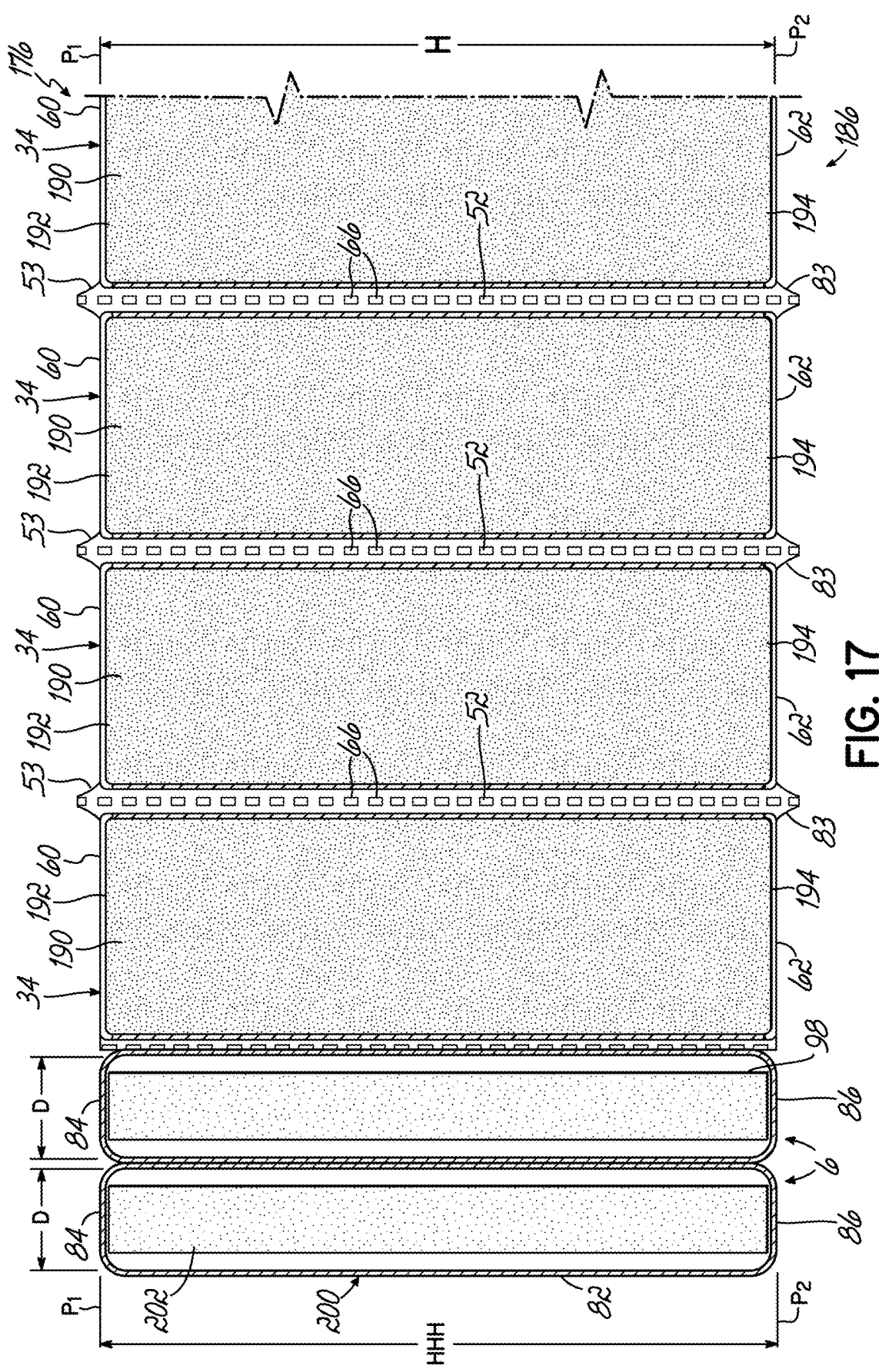


FIG. 16A



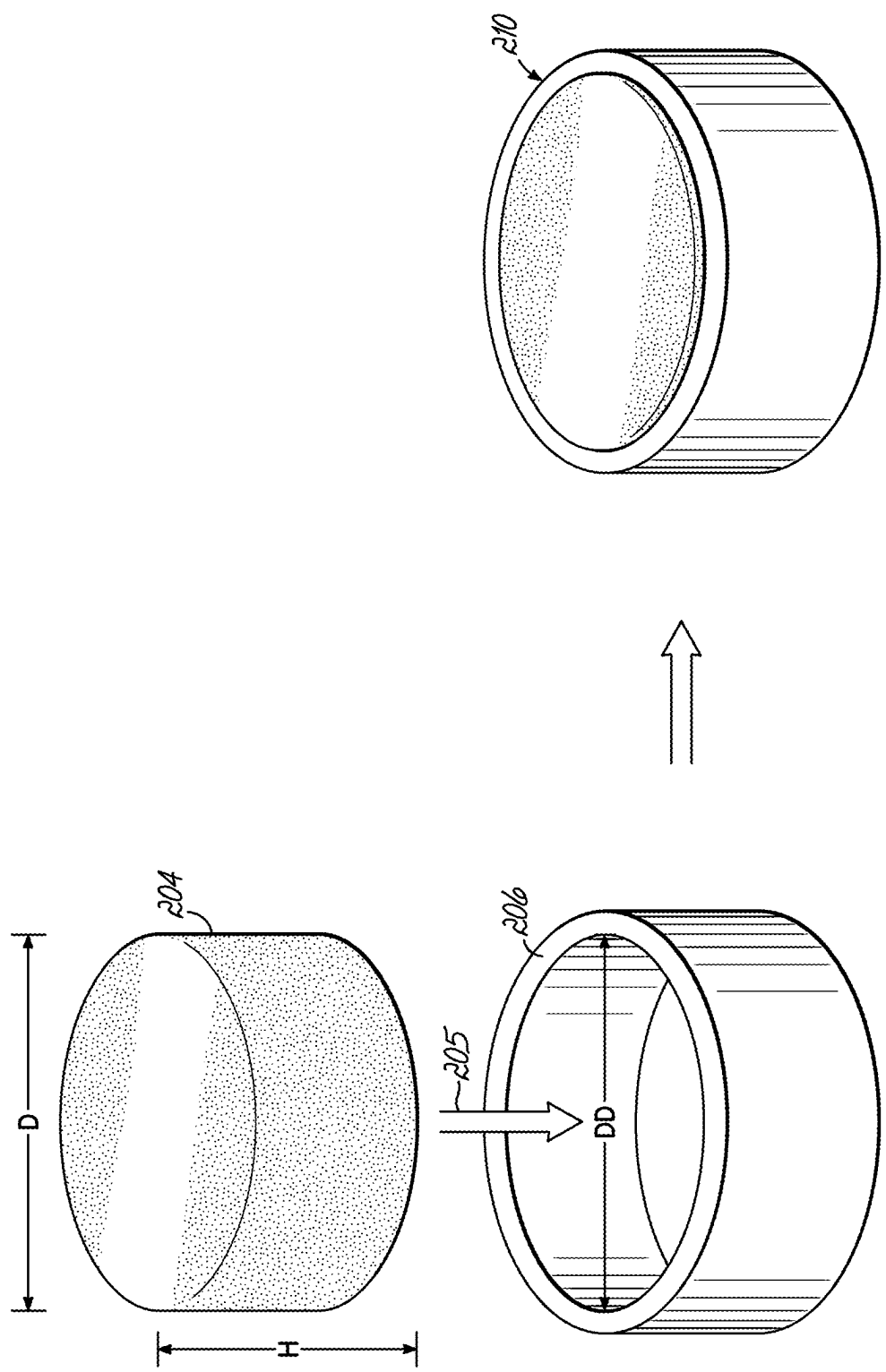


FIG. 18

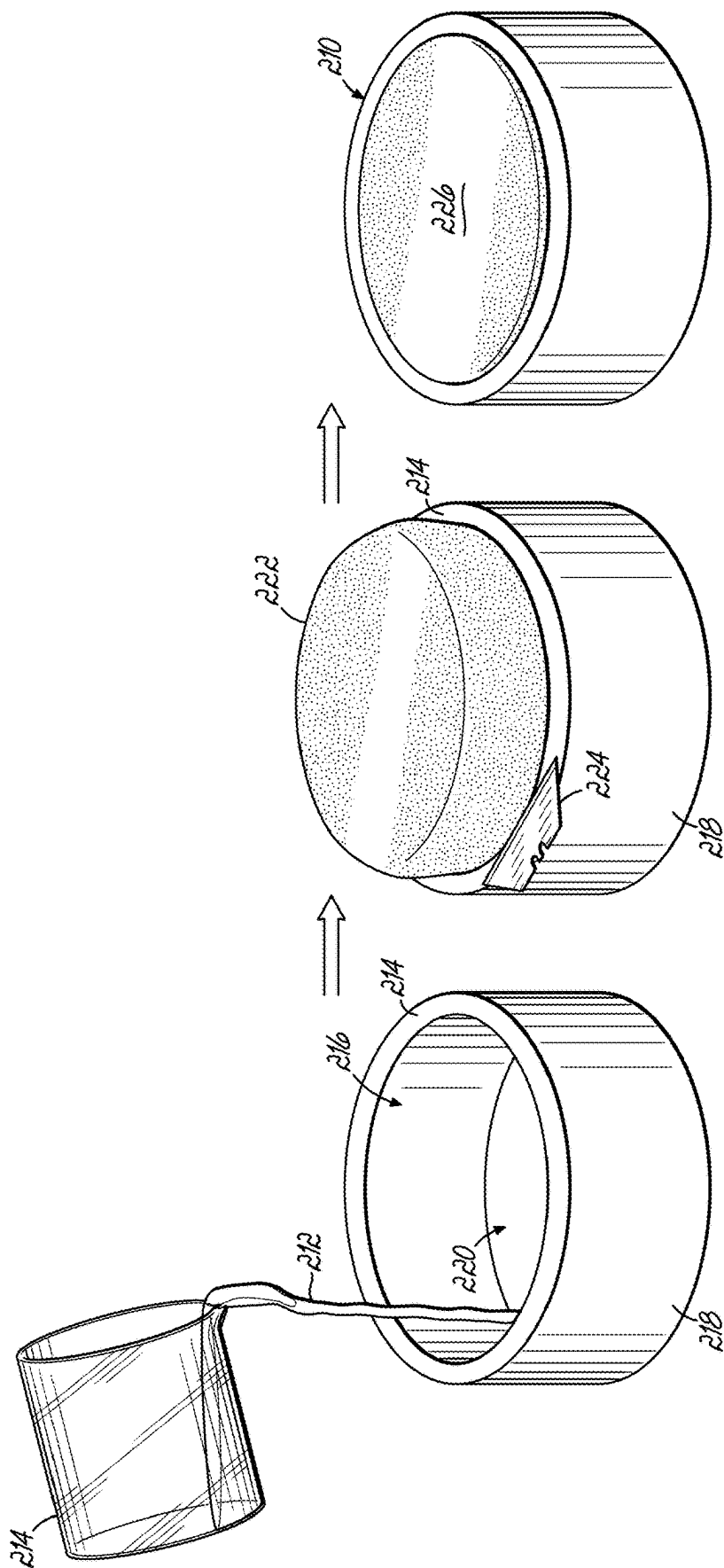
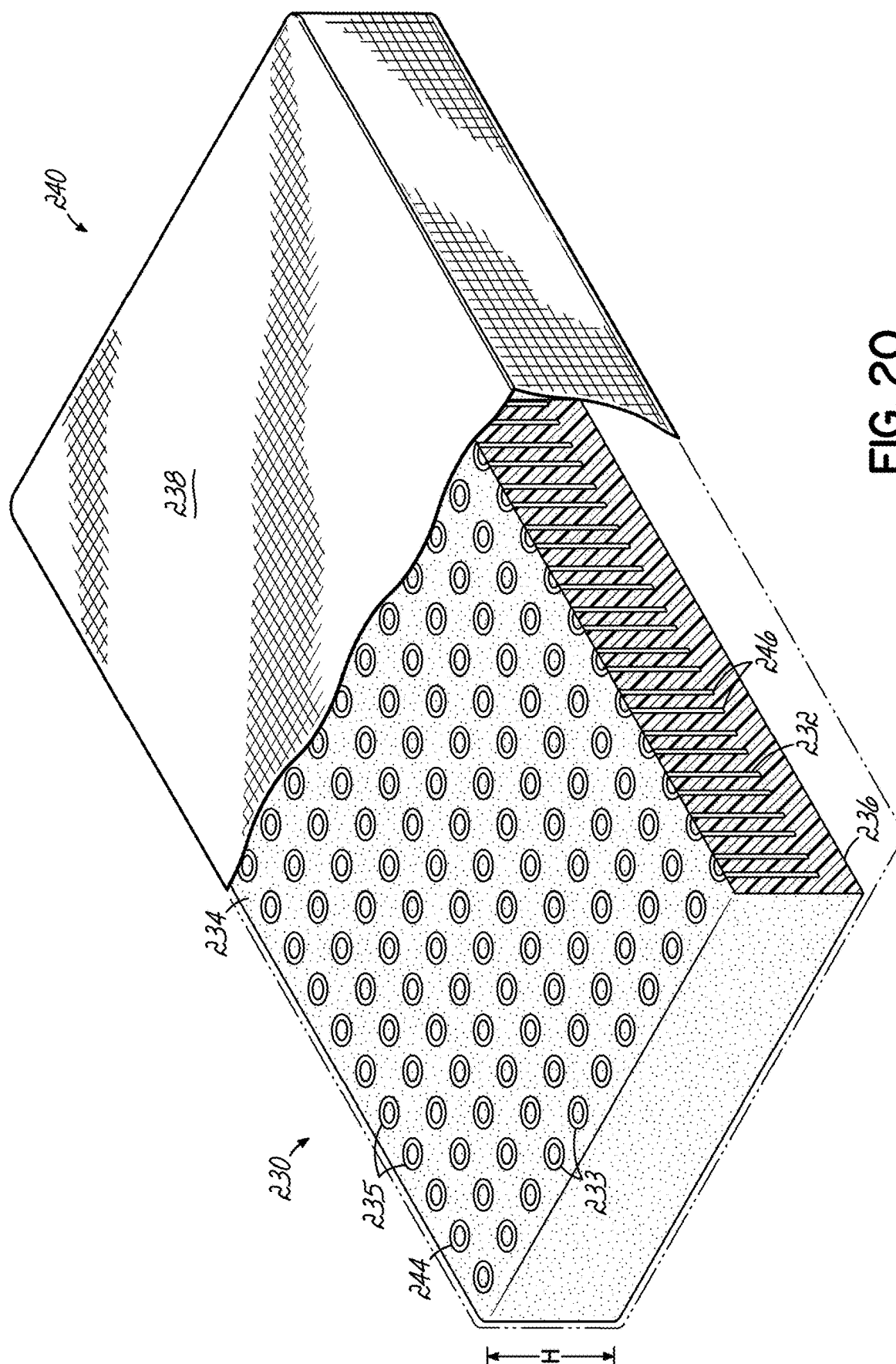


FIG. 19



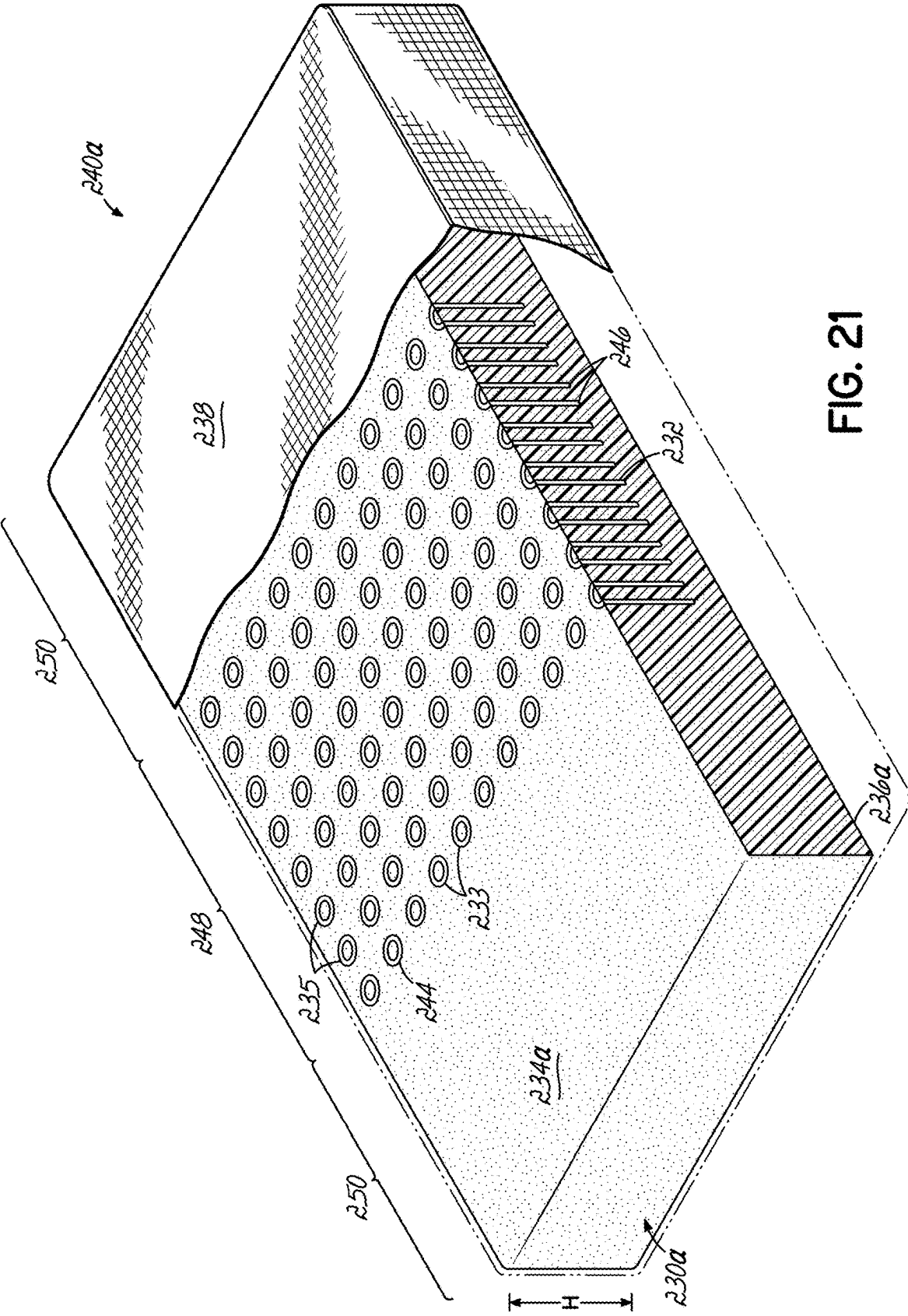


FIG. 21

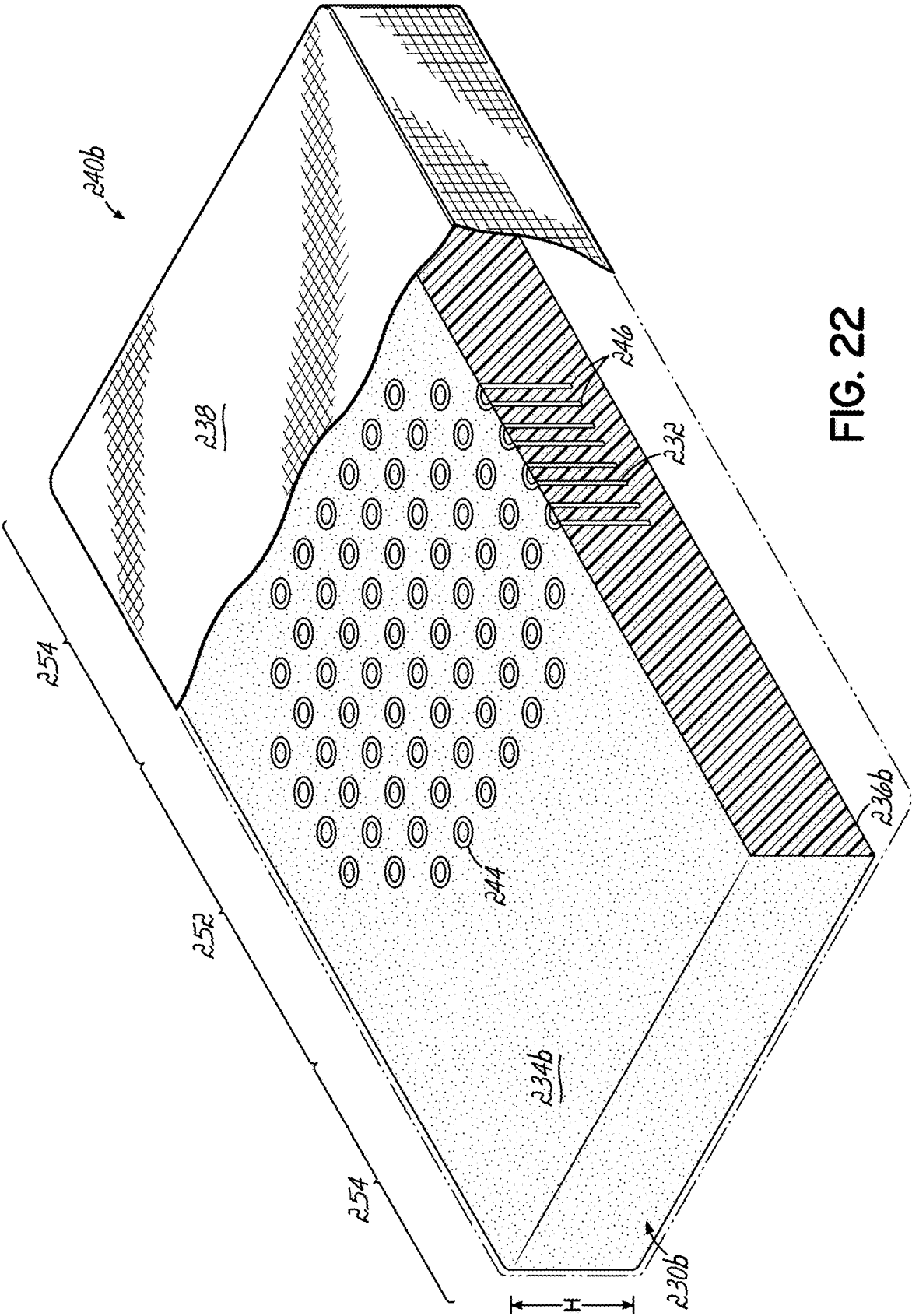
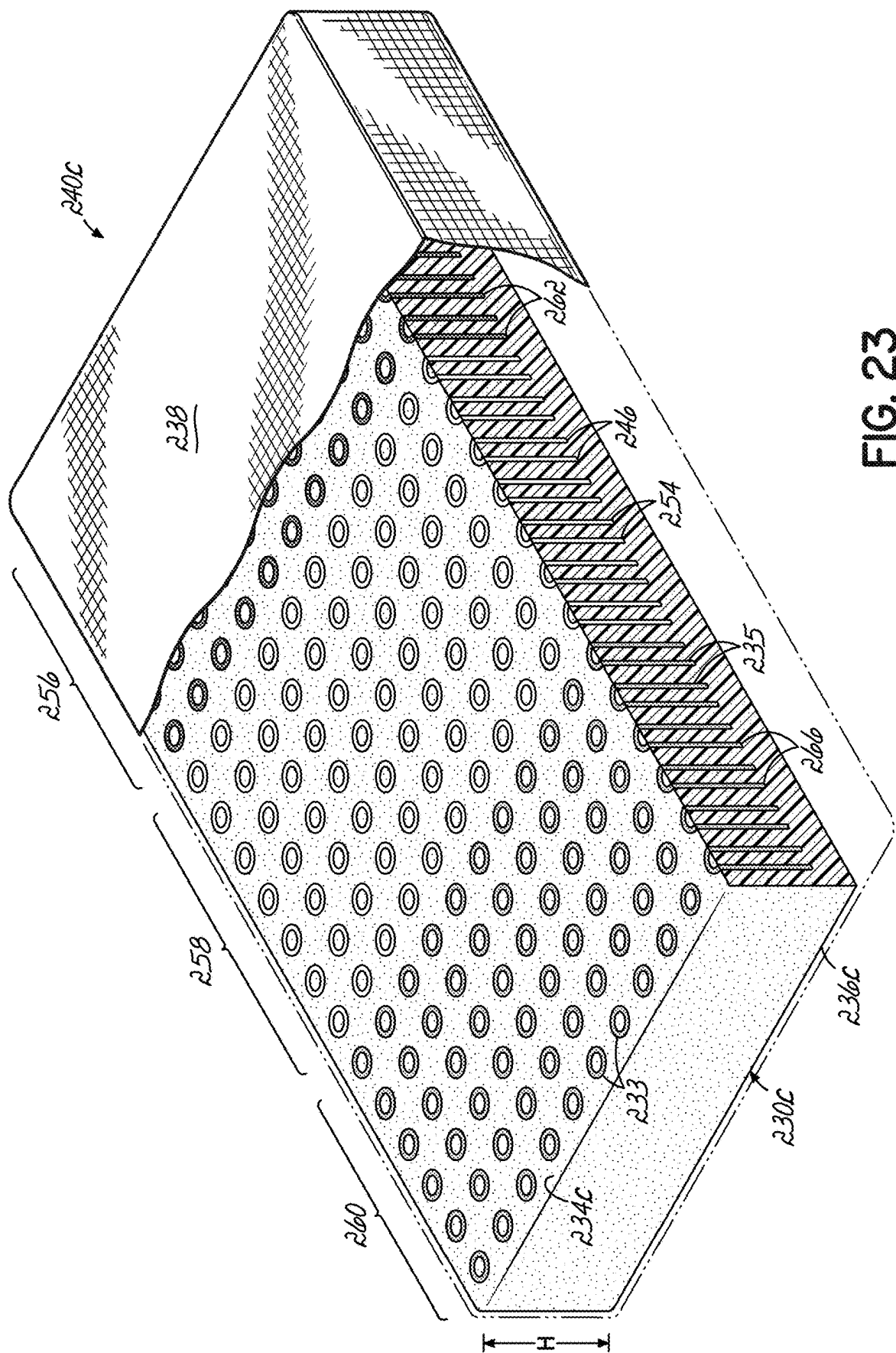


FIG. 22



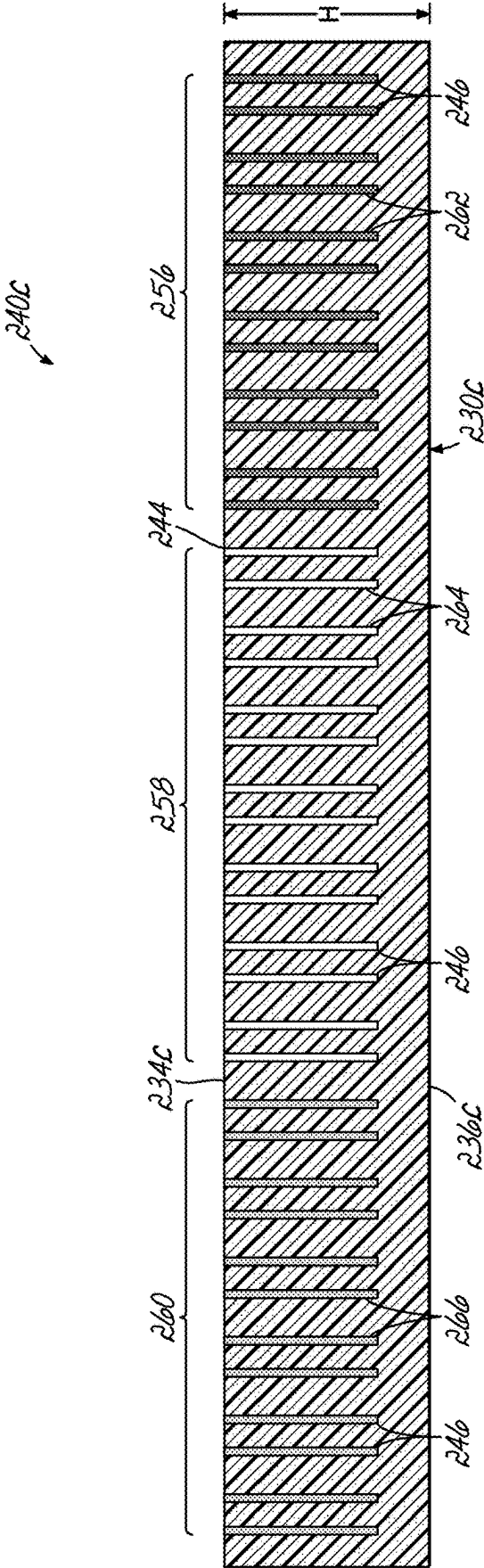
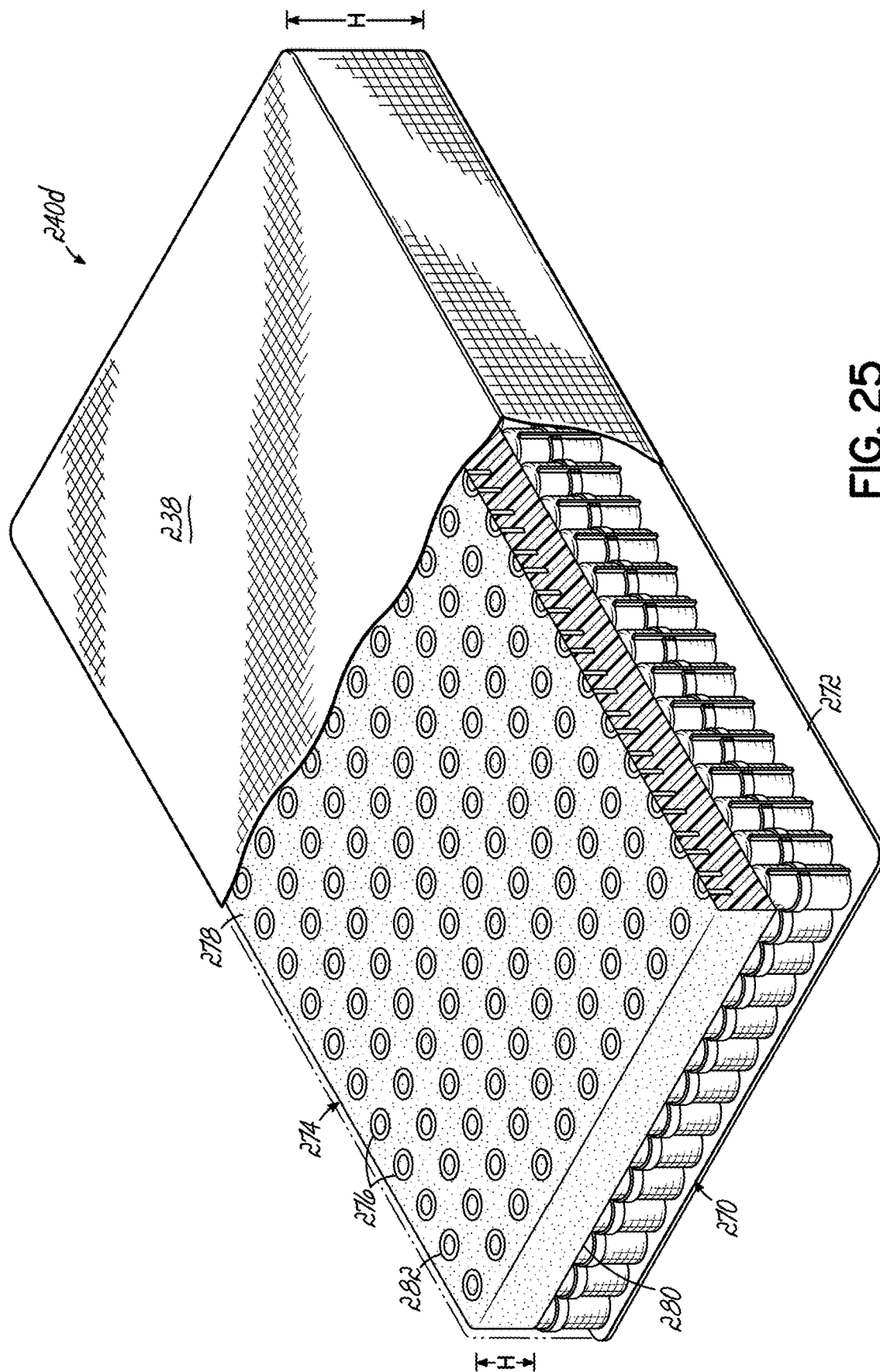


FIG. 24



POCKETED FOAM FILLED BUCKLING MEMBER ASSEMBLY

FIELD OF THE INVENTION

[0001] This invention relates generally to bedding and seating products and, more particularly, to pocketed buckling member assemblies used in bedding and seating products.

BACKGROUND OF THE INVENTION

[0002] Mattress spring core construction over the years has been a continuously improving art with advancements in materials and machine technology. A well-known form of spring core construction is known as a Marshall spring construction wherein metal coil springs are encapsulated in individual pockets of fabric and formed as elongate or continuous strings of pocketed coil springs. In an earlier form, these strings of coil springs were manufactured by folding an elongate piece of fabric in half lengthwise to form two plies of fabric and stitching transverse and longitudinal seams to join the plies of fabric to define pockets within which the springs were enveloped.

[0003] Improvements in spring core constructions have involved the use of fabrics which are thermally or ultrasonically weldable to themselves. By using such welding techniques, these fabrics have been advantageously used to create strings of individually pocketed coil springs wherein transverse and longitudinal welds, instead of stitching, are used to form the pockets encapsulating the springs.

[0004] Once strings of pocketed springs are constructed, they may be assembled to form a pocketed spring core or assembly for a mattress, cushion or the like by a variety of methods. For example, multiple or continuous strings may be arranged in a row pattern corresponding to the desired size and shape of a mattress or the like, and adjacent rows of strings may be interconnected by a variety of methods. The result is a unitary assembly of pocketed coil springs serving as a complete spring core assembly.

[0005] Conventional pocketed spring cores incorporating pocketed strings of springs have less motion transfer between sleeping partners when compared to traditional helically-laced open coil spring assemblies. Each pocketed coil spring moves with greater independence and, therefore provide less influence on adjacent pocketed coil springs than if the coil springs were not inside individual pockets. However, with a traditional pocketed spring mattress, a sheet of foam or other cushioning layer is attached to an upper surface of the pocketed spring assembly. The foam or cushioning sheet or sheets acts like a bridge, such that a load applied to one side of a mattress affects the other side of the mattress, providing an undesirable bridging effect. The present invention eliminates the undesirable bridging effect by encapsulating individual cushion members inside outer pockets of strings of springs.

[0006] U.S. Pat. No. 11,229,299 discloses a pocketed spring assembly incorporating strings which have individually pocketed coil springs with a cushioning pad and a buckling member inside the outer pockets. However, due to the inner coil springs being individually pocketed and contained within an outer pocket with a cushioning pad and a buckling member, the manufacture of such a product may be expensive.

[0007] Therefore, there remains a need to combine multiple technologies to reduce the manufacturing costs of a bedding or seating product having buckling members inside pockets.

[0008] There is further a need for a buckling member having additives to provide desired properties to a pocketed buckling member layer for use in a bedding or seating product.

SUMMARY OF THE INVENTION

[0009] In one aspect, a bedding or seating product is provided. The product comprises a pocketed buckling member assembly comprising a plurality of parallel strings of pocketed buckling members, each string being joined to at least one adjacent string. Each string comprises a piece of fabric surrounding a plurality of buckling members, first and second opposed plies of fabric being on opposite sides of the buckling members. A plurality of pockets having the same height are formed along the length of the string by a longitudinal seam and transverse seams joining the first and second plies. A single buckling member is in each of the pockets. Each buckling member has a height approximately equal to the height of the pocket in which the buckling member resides. Each of the buckling members may be a unitary member having an outer wall. These buckling members may have an open top and open bottom. The buckling members may have any number of passages of any desired size or shape.

[0010] Each of the buckling members is made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythioethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof.

[0011] According to some embodiments, at least some of the buckling members are at least partially filled with foam. The foam may be any known foam including, but limited to, polyurethane or latex foam with or without additives.

[0012] Cushioning materials may be placed on the pocketed buckling member assembly, and a covering, usually an upholstered covering, encases the pocketed buckling member assembly and cushioning materials.

[0013] The strings may extend longitudinally (from end-to-end) or transversely (from side-to-side). A pocketed buckling member assembly for use in a bedding or seating product may be posturized into regions or zones of different firmness by incorporating different strings into the pocketed buckling member assembly.

[0014] At least some of the buckling members may contain at least one of the following additives: phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes, density reducing agents or gas generating materials. Such additives may impart different properties to the buckling members such as firmness, for example. The additives may be in the form of conductive solids, powders, particles or flakes.

[0015] At least some of the buckling members may have an outer wall and a hollow interior without any internal ribs

or center. These buckling members may have an open top and open bottom which define one passage. The passage may be at least partially filled with foam.

[0016] In another aspect, a pocketed buckling member assembly for a bedding or seating product is provided. The pocketed buckling member assembly comprises a plurality of parallel strings of pocketed buckling members. Each string is joined to at least one adjacent string. Each of the strings comprises a plurality of interconnected pockets made from one piece of fabric. Each of the pockets contains one buckling member encased in fabric, the buckling member filling the pocket with nothing else inside the pocket. The piece of fabric is joined to itself along a longitudinal seam and has first and second opposed plies of fabric on opposite sides of the buckling members. The first and second plies of fabric are joined by transverse seams between adjacent buckling members creating individual pockets, each pocket containing a single buckling member. Each of the buckling members is made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof.

[0017] At least some of the buckling members may contain at least one of the following additives: phase change materials, plant-based fibers such as cellulose, cotton, hemp, linen, jute, wool and the like, synthetic fibers including carbon fiber, nylon, olefin, acrylic, polyester, rayon, vinyon, spandex, vinalon, aramids, Modal, PBI, Orlon and the like, fragrances including slow-release fragrances, organic and inorganic antimicrobial additives such as zinc omadine, N-Butyl-1,2-Benzisothiazolin-3-one, OIT (2-n-Octyl-4-isothiazolin-3-one, 3-(Trimethoxysilyl) propyldimethyloctadecyl ammonium chloride, copper and silver and the like, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes such as activated carbon, carbon nanotubes, graphene, graphite, aluminum powder, silicon carbide, diamond dust and the like may be incorporated into the material. Density reducing agents such as fumed silica or gas generating materials may be added too. At least some of the buckling members may be filled with foam.

[0018] In some embodiments, each of the buckling members is a unitary member having an outer wall, internal ribs extending inwardly from the outer wall to a center, an open top and an open bottom defining passages between the internal ribs. In some of these embodiments, the buckling member has at least three internal ribs.

[0019] In another aspect, a pocketed buckling member assembly comprises a plurality of parallel strings of pocketed buckling members. Each string is joined to an adjacent string. Each of the strings comprises a plurality of interconnected outer pockets made from one piece of fabric. Each of the pockets is approximately the same height. Each pocket contains a single buckling member having the same height as the pocket. The piece of fabric is joined to itself along a longitudinal seam and has first and second opposed plies of fabric on opposite sides of the buckling members. The fabric of the first and second plies is joined by transverse seams. Some of the strings of pocketed buckling members have buckling members made of a different material than the buckling members of other strings. However, the buckling

members within a string are the same or substantially the same. Such a pocketed buckling member assembly may have edge support due to the buckling members of the outermost strings being different than the buckling members of the interior strings.

[0020] Each of the buckling members may be a unitary member having a hollow interior and being made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythioethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof.

[0021] At least some of the buckling members contain at least one of the following additives: phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes, density reducing agents or gas generating materials.

[0022] According to some embodiments, at least some of the buckling members are at least partially filled with foam. The foam may be any known foam including, but limited to, polyurethane or latex foam with or without additives.

[0023] In any embodiment shown or described herein, at least some of the buckling members may be a unitary member having an outer wall, a center, and internal ribs extending between the outer wall and center. These buckling members may have an open top and open bottom which define passages between the internal ribs. In some embodiments, each of the buckling members may have at least three internal ribs. The buckling members may have any number of passages of any desired size or shape.

[0024] One advantage of the present invention is that when a bedding or seating product, such as a mattress, is manufactured, the manufacturer need not purchase metal coil springs, which may be expensive.

[0025] Another advantage of the present invention is that the buckling members may be individually pocketed after having been formed with additives. The additives may impart a cooler “feel” to the user of the bedding or seating product in some areas or regions of a bedding or seating product when compared to other areas or regions.

[0026] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the summary of the invention given above, and the detailed description of the drawings given below, serve to explain the principles of the present invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 is a perspective view, partially broken away, of a bedding or seating product incorporating a pocketed buckling member assembly surrounded by a border.

[0028] FIG. 1A is a perspective view, partially broken away, of a bedding or seating product incorporating another pocketed buckling member assembly surrounded by a border.

[0029] FIG. 1B is a perspective view, partially broken away, of a double-sided bedding product incorporating another pocketed buckling member assembly surrounded by a border.

[0030] FIG. 1C is a perspective view, partially broken away, of another bedding product incorporating the pocketed buckling member assembly of FIG. 1 surrounded by a border.

[0031] FIG. 1D is a perspective view, partially broken away, of another bedding product incorporating the pocketed buckling member assembly of FIG. 1A.

[0032] FIG. 1E is a perspective view, partially broken away, of another bedding product incorporating the pocketed buckling member assembly of FIG. 1D surrounded by a foam border.

[0033] FIG. 2 is a perspective view, partially broken away, of a portion of a string of pocketed buckling members of the pocketed buckling member assembly of FIG. 1 in an unloaded condition.

[0034] FIG. 3 is a perspective view of a buckling member in a relaxed or unloaded condition.

[0035] FIG. 3A is a perspective view of the circled area 3A of FIG. 3 showing an additive incorporated into the buckling member.

[0036] FIG. 4 is a cross-sectional view, partially broken away, of a portion of the string of FIG. 2 in an unloaded condition.

[0037] FIG. 4A is a cross-sectional view, partially broken away, of a portion of the string of FIG. 2 in an unloaded condition and a portion of a border comprising pocketed springs.

[0038] FIG. 4B is a cross-sectional view, partially broken away, of a portion of the string of FIG. 2 in an unloaded condition and a portion of a border comprising pocketed buckling members.

[0039] FIG. 5 is a perspective view of a portion of the pocketed buckling member assembly of FIG. 1 in a relaxed condition.

[0040] FIG. 6 is a perspective view of a portion of another pocketed buckling member assembly in a relaxed condition, the strings being offset from one another.

[0041] FIG. 7A is a top view of an alternative buckling member.

[0042] FIG. 7B is a top view of an alternative buckling member.

[0043] FIG. 7C is a top view of an alternative buckling member.

[0044] FIG. 7D is a top view of an alternative buckling member.

[0045] FIG. 7E is a top view of an alternative buckling member.

[0046] FIG. 7F is a top view of an alternative buckling member.

[0047] FIG. 7G is a top view of an alternative buckling member.

[0048] FIG. 7H is a top view of an alternative buckling member.

[0049] FIG. 7I is a top view of an alternative buckling member.

[0050] FIG. 7J is a top view of an alternative buckling member.

[0051] FIG. 7K is a top view of an alternative buckling member.

[0052] FIG. 7L is a top view of an alternative buckling member.

[0053] FIG. 7M is a top view of an alternative buckling member.

[0054] FIG. 7N is a top view of an alternative buckling member.

[0055] FIG. 8 is a top view of a posturized pocketed buckling member assembly.

[0056] FIG. 9 is a top view of another posturized pocketed buckling member assembly.

[0057] FIG. 10 is a perspective view, partially broken away, of another bedding product incorporating a foam core and having a topper comprising a mini pocketed buckling member assembly.

[0058] FIG. 10A is a perspective view, partially broken away, of a portion of one of the strings of the topper of FIG. 10.

[0059] FIG. 10B is a perspective view of a buckling member used in the topper of FIG. 10 in a relaxed or unloaded condition.

[0060] FIG. 11 is a perspective view, partially broken away, of a bedding product like FIG. 10 but having a posturized topper comprising a mini pocketed buckling member assembly.

[0061] FIG. 12 is a cross-sectional view, partially broken away, of another bedding product incorporating a foam core and having a topper comprising a pocketed buckling member assembly and foam topper pieces.

[0062] FIG. 13 is an enlarged perspective view of a portion of a comfort layer partially disassembled and a showing a portion of a welding tool.

[0063] FIG. 14 is an enlarged perspective view of a portion of another comfort layer partially disassembled and a showing a portion of another welding tool.

[0064] FIG. 15 is a perspective view of several different shapes of buckling members filled with foam.

[0065] FIG. 16 is a perspective view, partially broken away, of a portion of a string of pocketed buckling members filled with foam of a pocketed buckling member assembly in an unloaded condition.

[0066] FIG. 16A is a perspective view of a portion of a pocketed foam filled buckling member assembly in a relaxed condition.

[0067] FIG. 17 is a cross-sectional view, partially broken away, of a portion of a string in an unloaded condition and a portion of a border comprising pocketed buckling members filled with different foams.

[0068] FIG. 18 is a schematic view of a method of constructing a foam filled buckling member.

[0069] FIG. 19 is a schematic view of another method of constructing a foam filled buckling member.

[0070] FIG. 20 is an isometric view of a foam mattress including buckling members therein.

[0071] FIG. 21 is an isometric view of a posturized foam mattress having buckling members therein.

[0072] FIG. 22 is an isometric view of a posturized foam mattress having buckling members therein.

[0073] FIG. 23 is an isometric view of a posturized foam mattress having buckling members therein.

[0074] FIG. 24 is a cross-sectional view of the posturized foam mattress of FIG. 23 without the covering.

[0075] FIG. 25 is an isometric view of a mattress having a posturized foam layer having buckling members therein above a pocketed spring assembly.

DETAILED DESCRIPTION OF THE INVENTION

[0076] Referring first to FIG. 1, there is illustrated a bedding product in the form of a single-sided mattress 10 incorporating the principles of the present invention. This product or mattress 10 comprises a pocketed buckling member assembly 12 over the top of which there lay conventional padding or cushioning layers 14, 16 which may be foam, fiber, gel, a pocketed spring blanket or any other suitable materials or any combination thereof. The pocketed buckling member assembly 12 is mounted upon a base 18 and is completely enclosed within an upholstered covering 20. The base 18 may be any known material including a flexible sheet such as a scrim sheet or a more rigid dimensionally stabilizing substrate as described in U.S. Pat. Nos. 11,013,340 or 11,771,235, which are each incorporated by reference herein.

[0077] The pocketed buckling member assembly 12 is surrounded by a border 17 comprising longitudinally extending border strings 5 of pocketed members 11 and transversely extending border strings 6 of pocketed members 11. The pocketed members 11 of the border 17 may be of a different diameter (usually smaller) than the diameter of the pocketed buckling members 28 of the pocketed buckling member assembly 12 to increase firmness along at least two edges or sides of the product. The pocketed members 11 of the border 17 may be border coil springs 96 of a diameter “D”, as shown in FIG. 4A, or border buckling members 98 of a diameter “D”, as shown in FIG. 4B.

[0078] Although one type of border 17 is illustrated, the border 17 may assume other forms or shapes of any desired size, such as pocketed border buckling members 98 of different geometries, different compositions, different sizes and/or different properties than the pocketed buckling members 28 of any of the pocketed buckling member assemblies disclosed herein. Alternatively, the border 17 may be omitted in this embodiment or any embodiment described or shown herein.

[0079] As shown in FIG. 1, fully assembled, the product 10 has a length “L” defined as the linear distance between opposed end surfaces 22 (only one being shown in FIG. 1). Similarly, the assembled product 10 has a width “W” defined as the linear distance between opposed side surfaces 24 (only one being shown in FIG. 1). In the product shown in FIG. 1, the length is illustrated as being greater than the width. However, it is within the scope of the present invention that the length and width may be identical, as in a square product.

[0080] As shown in FIG. 1, the pocketed buckling member assembly 12 is manufactured from multiple strings 26 of pocketed buckling members 28 joined together. Each string 26 of pocketed buckling members 28 extends longitudinally or from head-to-foot along the full length of the product 10.

[0081] Although the strings 26 of the pocketed buckling member assembly 12 are illustrated as extending longitudinally or from head-to-foot in the pocketed buckling member assembly 12 of FIG. 1, they may extend transversely or from side-to-side as shown in the pocketed buckling member assembly 12a shown in the product 10a shown in FIG. 1A. The pocketed buckling member assembly 12a comprises multiple strings 26a of pocketed buckling members, identical to the strings 26, but shorter in length.

[0082] FIG. 1B illustrates a double-sided mattress 10b comprising a pocketed buckling member assembly 12 sur-

rounded with a border 17. However, the mattress 10b of FIG. 1B has conventional padding layers 14, 16 above and below a pocketed buckling member assembly 12. The pocketed buckling member assembly 12 comprises a plurality of strings 26 extending from head to foot or longitudinally. However, the transversely extending strings 26a of FIG. 1A could be used in a double-sided product including a mattress.

[0083] FIG. 1C illustrates a single-sided mattress 10c comprising a pocketed buckling member assembly 12 surrounded by a border 17. However, the mattress 10c of FIG. 1C has a pocketed topper 19 comprising pocketed mini coil springs in addition to padding layers 14, 16 above the pocketed topper 19. A scrim layer 21 separates the pocketed topper 19 from the pocketed buckling member assembly 12. Although one configuration of pocketed topper 19 is illustrated, any pocketed topper or comfort layer known in the art may be used.

[0084] FIG. 1D illustrates a single-sided mattress 10d comprising pocketed buckling member assembly 12a the same as in the mattress 10a of FIG. 1A. However, the mattress 10d of FIG. 1D has no border. Although the strings 26a of pocketed buckling member assembly 12a extend transversely or from side to side, they may extend longitudinally or from head to foot without a border in any of the embodiments shown or described herein.

[0085] FIG. 1E illustrates a single-sided mattress 10e comprising pocketed buckling member assembly 12 the same as in the mattress 10 of FIG. 1. However, the mattress 10e of FIG. 1E has a foam border 15. Although the strings 26 of pocketed buckling member assembly 12 extend longitudinally or from head to foot, they may extend transversely or from side to side in any of the embodiments shown or described herein.

[0086] According to this invention, any of the padding or cushioning layers, including the pocketed topper 19, may be omitted in any of the embodiments shown or described herein.

[0087] These strings of pocketed buckling members 26 and 26a, and any other strings described or shown herein, may be connected in side-by-side relationship as, for example, by gluing the sides of the strings together in an assembly machine, to create an assembly or matrix of springs having multiple rows and members of pocketed buckling members bound together as by gluing, welding or any other conventional assembly process commonly used to create pocketed spring cores or assemblies.

[0088] Referring to FIGS. 5 and 6, the strings 26 of pocketed buckling members may be joined so that the individually pocketed buckling members 28 are aligned in transversely extending rows 30 and longitudinally extending members 32. Alternatively, the strings 26 may be offset from one another in a pocketed buckling member assembly. In such an arrangement, shown in FIG. 6, the individually pocketed buckling members 28 are not aligned in rows and members; instead the individually pocketed buckling members 28 fill gaps or voids 70 of the adjacent strings 26. FIG. 6 shows a portion of a pocketed buckling member assembly 12' with multiple strings 26 arranged in this manner. Either alignment of strings may be incorporated into any of the pocketed buckling member assemblies or cores illustrated or described herein. Although FIGS. 5 and 6 illustrate strings 26, the same alignments may be used in any pocketed

buckling member assembly disclosed herein having any strings disclosed herein including strings 26a.

[0089] FIG. 2 illustrates a perspective view of the portion of a string 26 of pocketed buckling members 28 in a relaxed condition under no external load. As best illustrated in FIG. 2, each string 26 of pocketed buckling members 28 comprises a row of interconnected fabric pockets 34. Each of the fabric pockets 34 contains a single buckling member 36 and no other items. In other words, each of the fabric pockets 34 contains solely a single buckling member 36.

[0090] Preferably, one piece of fabric is used to create the pockets 34 of the string 26 of pocketed buckling members 28, the piece of fabric being folded over onto itself around the buckling members 28. As best shown in FIG. 2, opposite sides or plies 47, 49 of the fabric are sewn, welded or otherwise secured together to create a longitudinal seam 50 and a plurality of separating or transverse seams 52. FIG. 2 illustrates ply 47 being closest to the reader and ply 49 being behind the buckling members 28.

[0091] Although the seams or welds in the embodiments shown in this document are shown as being welded spaced rectangles, any of the seams may be spaced dots, triangles or solid line segments with or without spaces. The drawings are not intended to be limiting.

[0092] As best shown in FIG. 2, opposed edges 56 of the piece of fabric used to create the string 26 of pocketed buckling members 28 are aligned and spaced from the longitudinal seam 50 a distance indicated by numeral 58. Although the drawings indicated the longitudinal seam 50 being below the free edges 56 of the piece of fabric, the longitudinal seam 50 may be above the free edges 56 of the piece of fabric.

[0093] As shown in FIGS. 2, 4A and 4B, the piece of fabric used to create the string 26 has a plurality of upper ears 53 and a plurality of lower ears 83. When the ears are collapsed, the string 26 has a generally planar top surface 60 in a top plane P1 and a parallel generally planar bottom surface 62 in a bottom plane P2. The linear distance between the top and bottom surfaces 60, 62 of the string 26 defines a height "H" of the string 26. This linear distance further defines the height H of the pocketed buckling member assembly 12 because each of the strings 26 has the same height. However, it is within the scope of the present invention that different strings of a pocketed buckling member assembly have different heights.

[0094] As shown in FIGS. 4A and 4B, in one embodiment, the transverse seams 52 of string 26 separating adjacent pockets extend from the top of an upper ear 53 of fabric to the bottom of a lower ear 83 of fabric. However, the transverse seams 52 may be any desired length.

[0095] An ultrasonically weldable fabric is used to create the string of pocketed buckling members 26. The ultrasonically weldable fabric is permeable to airflow through the fabric itself due to the nature of the fabric. Air moves between adjacent fabric pockets 38 and into and out of the string 26 through the ultrasonically weldable fabric.

[0096] As best shown in FIG. 2, the longitudinal seam 52 comprises multiple spaced linear weld segments 64 formed using an ultrasonic welding horn and anvil (not shown) as known in the art. At least some of the longitudinal seams 52 of a string may not be segmented or be only partially segmented. For example, the longitudinal seam 52 of a string may not be segmented at all.

[0097] As best shown in FIGS. 4A and 4B, each transverse or separating seam 52 comprises multiple spaced linear weld segments 66 formed using an ultrasonic welding horn and anvil (not shown) to join the opposed sides or plies 47, 49 of the ultrasonically weldable fabric. Again, at least some of the transverse or separating seams 52 of a string may not be segmented or may be only partially segmented. For example, one or more transverse seams 52 of a string may be partially segmented or not be segmented at all. Although the weld segments in the embodiments shown are illustrated as being heat-welded spaced rectangular-shaped segments, any of the seam segments may be other shapes, such as spaced dots, ovals or triangles of any desired sizes.

[0098] As best shown in FIGS. 2 and 3, the buckling member 36 is shown as being cylindrically shaped member like a can. Buckling member 36 is illustrated having an outer wall 38 having an outer surface 40 and an inner surface 42, the distance between which defines the thickness "T" of the outer wall 38. The buckling member 36 has three internal ribs 44 which extend from the outer wall 38 to a center 46. The buckling member 36 has a generally planar upper surface 67 and a generally planar lower surface 69 which define the height "HH" of the buckling member 36. The buckling member 36 has an open top 71 and open bottom 73 defining three passages 75 between the internal ribs 44. Although the drawings illustrate one thickness "T" of outer wall 38, the outer wall may be any desired thickness. Although the drawings illustrate the internal ribs 44 having one thickness, the internal ribs may be any desired thickness.

[0099] In any embodiment shown or described herein, any of the buckling members 36; 36a-36h, 36i or 36j shown or described herein can be made of acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof.

[0100] Regardless of the material used to make the buckling member 36, additives 80 such as slow-release fragrances, antimicrobial additives such as copper and silver, color additives, thermochromic materials, bed bug inhibitors, conductive additives such as graphite, aluminum powder, silicon carbide and diamond dust may be incorporated into the material. Density reducing agents such as fumed silica or gas generating materials may be added too. FIG. 3A illustrates additives 80 incorporated into the material of the buckling member 36.

[0101] As shown in FIGS. 2 and 3, the piece of fabric used to create the string 26 has a plurality of upper ears 53 and a plurality of lower ears 83. When the ears are collapsed, the string 26 has a generally planar top surface 60 in a top plane P1 and a parallel generally planar bottom surface 62 in a bottom plane P2. The linear distance between the top and bottom surfaces 60, 62 of the string 26 defines a height "H" of the string 26. This linear distance further defines the height H of the pocketed buckling member assembly 12 because each of the strings 26 has the same height. However, it is within the scope of the present invention that different strings of a pocketed buckling member assembly have different heights.

[0102] As shown in FIGS. 2 and 3, in one embodiment, the transverse seams 52 of string 26 separating adjacent pockets extend from the top of an upper ear 53 of fabric to the bottom of a lower ear 83 of fabric.

[0103] FIG. 4A illustrates one embodiment of mattress 10 having a pocketed buckling member assembly 12 and border strings 6 of pocketed border coil springs 96 extending perpendicular to the head-to-foot or longitudinal direction of the strings 26 of buckling members 36. Each of the border strings 6 extends from side-to-side or transversely. Each border string 6 comprises a single piece of fabric made into a plurality of individual border pockets 82 of the same height HHH defined as the linear distance between an upper surface 84 of the pocket in plane P1 and a lower surface 86 of the border pocket 82 located in plane P2. The height HHH of the border strings 6 is preferably identical to the height H of the pockets 34 of the pocketed buckling member assembly 12. However, these heights may be different. A coil spring 96 having a diameter D is located in each of the border pockets 82. Although one configuration of coil spring 96 is illustrated any known coil spring may be used.

[0104] Although not shown, each of the border strings 5 extending from head-to-foot or longitudinally may contain a coil spring 96 and may be any desired height.

[0105] FIG. 4B illustrates a view like FIG. 4A but showing border buckling members 88 in place of coil springs 96 in the border pockets 82. Each of the border buckling members 88 is shown having a diameter D, the same as the diameter of the coil springs 96. However, the border buckling members 88 may have any desired diameter.

[0106] FIGS. 7A-7H illustrate top views of different buckling members which may be substituted for the buckling members 36 shown and described herein. Although each of the buckling members shown in FIGS. 7A-7H are illustrated being un-pocketed any of them may be individually pocketed. Regardless of the material used to make any of the buckling members, additives 80 such as phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes, density reducing agents or gas generating materials may be incorporated into the material. Density reducing agents such as fumed silica or gas generating materials may be added too.

[0107] FIG. 7A illustrates a cross-sectional view of another cylindrical shaped buckling member 36a which may be used in any embodiment of string shown or described herein. Buckling member 36a is illustrated having a circular outer wall 38a having an outer surface 40a and an inner surface 42a, the distance between which defines the thickness “Ta” of the circular outer wall 38a. The buckling member 36a has four internal ribs 44a which extend from the outer wall 38a to a center 46a. The buckling member 36a may be any desired height. The buckling member 36a has an open top and open bottom defining four internal passages 75a between the internal ribs 44a and outer wall 38a. Although the drawings illustrate one thickness “Ta” of outer wall 38a, the outer wall 38a may be any desired thickness. Although the drawings illustrate the internal ribs 44a having one thickness, the internal ribs may be any desired thickness.

[0108] FIG. 7B illustrates a cross-sectional view of another cylindrical shaped buckling member 36b which may be used in any embodiment of string shown or described herein. Buckling member 36b is illustrated having a circular

outer wall 38b having an outer surface 40b and an inner surface 42b, the distance between which defines the thickness “Tb” of the circular outer wall 38b. The buckling member 36b has no internal ribs and may be any desired height. The buckling member 36b has an open top and open bottom defining one internal passage 75b inside outer wall 38b. Although the drawings illustrate one thickness “Tb” of outer wall 38b, the outer wall 38b may be any desired thickness.

[0109] FIG. 7C illustrates a cross-sectional view of another buckling member 36c which may be used in any embodiment of string shown or described herein. Buckling member 36c is illustrated having a triangular shaped outer wall 38c having an outer surface 40c and an inner surface 42c, the distance between which defines the thickness “Tc” of the triangular outer wall 38c. The buckling member 36c may be any desired height. The buckling member 36c has an open top and open bottom defining one internal passage 75c inside outer wall 38c. Although the drawings illustrate one thickness “Tc” of outer wall 38c, the outer wall 38c may be any desired thickness.

[0110] FIG. 7D illustrates a cross-sectional view of another buckling member 36d which may be used in any embodiment of string shown or described herein. Buckling member 36d is illustrated having a square shaped outer wall 38d having an outer surface 40d and an inner surface 42d, the distance between which defines the thickness “Td” of the square outer wall 38d. The buckling member 36d may be any desired height. The buckling member 36d has an open top and open bottom defining one internal passage 75d inside outer wall 38d. Although the drawings illustrate one thickness “Td” of outer wall 38d, the outer wall 38d may be any desired thickness.

[0111] FIG. 7E illustrates a cross-sectional view of another buckling member 36e which may be used in any embodiment of string shown or described herein. Buckling member 36e is illustrated having a pentagon shaped outer wall 38e having an outer surface 40e and an inner surface 42e, the distance between which defines the thickness “Te” of the pentagon outer wall 38e. The buckling member 36e may be any desired height. The buckling member 36e has an open top and open bottom defining one internal passage 75e inside outer wall 38e. Although the drawings illustrate one thickness “Te” of outer wall 38e, the outer wall 38e may be any desired thickness.

[0112] FIG. 7F illustrates a cross-sectional view of another buckling member 36f which may be used in any embodiment of string shown or described herein. Buckling member 36f is illustrated having a hexagon shaped outer wall 38f having an outer surface 40f and an inner surface 42f, the distance between which defines the thickness “Tf” of the hexagon outer wall 38f. The buckling member 36f may be any desired height. The buckling member 36f has an open top and open bottom defining one internal passage 75f inside outer wall 38f. Although the drawings illustrate one thickness “Tf” of outer wall 38f, the outer wall 38f may be any desired thickness.

[0113] FIG. 7G illustrates a cross-sectional view of another buckling member 36g which may be used in any embodiment of string shown or described herein. Buckling member 36g is illustrated having a heptagon shaped outer wall 38g having an outer surface 40g and an inner surface 42g, the distance between which defines the thickness “Tg” of the heptagon outer wall 38g. The buckling member 36g

may be any desired height. The buckling member 36g has an open top and open bottom defining one internal passage 75g inside outer wall 38g. Although the drawings illustrate one thickness “Tg” of outer wall 38g, the outer wall 38g may be any desired thickness.

[0114] FIG. 7H illustrates a cross-sectional view of another buckling member 36h which may be used in any embodiment of string shown or described herein. Buckling member 36h is illustrated having an octagon shaped outer wall 38h having an outer surface 40h and an inner surface 42h, the distance between which defines the thickness “Th” of the pentagon outer wall 38h. The buckling member 36h may be any desired height. The buckling member 36h has an open top and open bottom defining one internal passage 75h inside outer wall 38h. Although the drawings illustrate one thickness “Th” of outer wall 38h, the outer wall 38h may be any desired thickness.

[0115] FIG. 7I illustrates a cross-sectional view of another buckling member 36cc which may be used in any embodiment of string shown or described herein. Buckling member 36cc is illustrated having the same configuration and structure as buckling member 36c but additionally has internal ribs 44c. Each of the internal ribs 44c extends from the outer wall 38c and converge at a center 46c. Although three internal ribs 44c are illustrated, any number of additional ribs may be added.

[0116] FIG. 7J illustrates a cross-sectional view of another buckling member 36dd which may be used in any embodiment of string shown or described herein. Buckling member 36dd is illustrated having the same configuration and structure as buckling member 36d but additionally has internal ribs 44d. Each of the internal ribs 44d extends from the outer wall 38d and ends at a center 46d. Although four internal ribs 44d are illustrated, any number of additional ribs may be added.

[0117] FIG. 7K illustrates a cross-sectional view of another buckling member 36ee which may be used in any embodiment of string shown or described herein. Buckling member 36ee is illustrated having the same configuration and structure as buckling member 36e but additionally has internal ribs 44e. Each of the internal ribs 44e extends from the outer wall 38e and ends at a center 46e. Although five internal ribs 44e are illustrated, any number of additional ribs may be added.

[0118] FIG. 7L illustrates a cross-sectional view of another buckling member 36ff which may be used in any embodiment of string shown or described herein. Buckling member 36ff is illustrated having the same configuration and structure as buckling member 36f but additionally has internal ribs 44f. Each of the internal ribs 44f extends from the outer wall 38f and ends at a center 46f. Although five internal ribs 44f are illustrated, any number of additional ribs may be added.

[0119] FIG. 7M illustrates a cross-sectional view of another buckling member 36gg which may be used in any embodiment of string shown or described herein. Buckling member 36gg is illustrated having the same configuration and structure as buckling member 36g but additionally has internal ribs 44g. Each of the internal ribs 44g extends from the outer wall 38g and ends at a center 46g. Although six internal ribs 44g are illustrated, any number of additional ribs may be added.

[0120] FIG. 7N illustrates a cross-sectional view of another buckling member 36hh which may be used in any

embodiment of string shown or described herein. Buckling member 36hh is illustrated having the same configuration and structure as buckling member 36h but additionally has internal ribs 44h. Each of the internal ribs 44h extends from the outer wall 38h and ends at a center 46h. Although eight internal ribs 44h are illustrated, any number of additional ribs may be added.

[0121] Referring now to FIG. 8, longitudinally extending strings are shown in one preferable arrangement for a pocketed buckling member assembly for a bedding or seating product, such as a mattress. As can be seen, the longitudinally extending strings are arranged in a plurality of zones on the pocketed buckling member assembly 12d. By way of example, two zones 72, 74 are illustrated, with the zones corresponding roughly to a “firm” side and a “soft” side. By way of further example, the longitudinally extending strings of the “soft” zone 72 may have a different buckling member 36 than the buckling members 36 of the longitudinally extending strings of the “firm” zone 74. Of course, other arrangements are within the scope of the invention. For example, a pocketed buckling member assembly like pocketed buckling member assembly 12d shown in FIG. 8 may comprise transversely extending strings rather than longitudinally extending strings. In such an arrangement, each transversely extending string would have to be half firm and half soft. In one example, each string would have different buckling members.

[0122] Referring now to FIG. 9, the transversely extending strings are shown in one preferable arrangement for a pocketed buckling member assembly 12e for a bedding or seating product, such as a mattress. As can be seen, the transversely extending strings are arranged in a plurality of zones on the pocketed buckling member assembly 12e. By way of example, three zones are illustrated, with the zones corresponding roughly to the location of a sleeper’s head and shoulders, mid-section, knees and feet. By way of further example, the two end “soft” zones 90 each comprise strings of springs having a softer buckling members than the buckling members of the transversely extending strings of the middle or “firm” zone 92 are strings. Of course, other arrangements are within the scope of the invention. For example, the mattress shown in FIG. 9 may comprise longitudinally extending strings rather than transversely extending strings. In such an arrangement, each longitudinally extending string would have to be divided into three sections; a middle “firm” section and two end or “soft” sections. Therefore, each string would have only the end thirds of the string having softer buckling members than the middle third.

[0123] FIGS. 10-10A illustrate an alternative embodiment of product such as a mattress 10f having a foam core 94 and a topper 112 residing on top of the foam core 94. The topper 112 comprises a mini pocketed buckling member assembly 102 which has a bottom scrim sheet 100 which may be glued or otherwise adhered to an upper surface 99 of the foam core 94. See FIG. 10A. The mini pocketed buckling member assembly 102 of topper 112 is illustrated comprising multiple strings 26f of pocketed buckling members 28f which are shorter than the strings 26 of the pocketed buckling members 28 described herein. As seen in FIG. 10, the strings 26f of pocketed buckling members 28f may be joined so that the individually pocketed buckling members 28f are aligned in transversely extending rows 30 and longitudinally extend-

ing members 32. Alternatively, the strings 26f may be offset from one another, as shown in FIG. 6.

[0124] Although FIGS. 10-10A illustrate a foam core 94, the core on which the topper 112 resides may be any conventional core including a pocketed spring core, a pocketed buckling member core or combination of known elements.

[0125] The foam core 94 and a topper 112 residing above the foam core 94 are completely enclosed within an upholstered covering 20. As shown in FIG. 10, fully assembled, the mattress 10f has a length “L” defined as the linear distance between opposed end surfaces 22 (only one being shown in FIG. 10). Similarly, the assembled mattress 10f has a width “W” defined as the linear distance between opposed side surfaces 24 (only one being shown in FIG. 10). In the product shown in FIG. 10, the length is illustrated as being greater than the width. However, it is within the scope of the present invention that the length and width may be identical, as in a square product.

[0126] FIG. 10A illustrates a perspective view of the portion of a string 26f of pocketed buckling members 28f in a relaxed condition under no external load. As best illustrated in FIG. 10A, each string 26f of pocketed buckling members 28f comprises a row of interconnected fabric pockets 34f. Each of the fabric pockets 34f contains a single buckling member 36f. For simplicity, like numbers are used for like parts.

[0127] The string 26f has a generally planar top surface 60f in a top plane P1 and a parallel generally planar bottom surface 62f in a bottom plane P2. The linear distance between the top and bottom surfaces 60f, 62f of the string 26f defines a height “HT” of the string 26f. This linear distance further defines the height HT of the topper 112 because the strings 26f are approximately the same height. However, it is within the scope of the present invention that different strings of springs of the topper have different heights.

[0128] Although the topper 112 is shown comprising longitudinally extending strings 26f of individually pocketed buckling members 36f, the strings may extend transversely. Although the height HT of topper 112 is shown in the drawings to be a certain distance, the height HT of topper 112 may be any desired height but is preferably less than the height of the foam core 94.

[0129] Although FIGS. 10-10A illustrate buckling members 36f shaped like buckling members 36 but being a lesser height, the buckling members used in any topper shown or described herein may be any of the buckling members shown or described herein.

[0130] FIG. 10B illustrates a buckling member 36tt shaped identical to buckling member 36f of FIGS. 10-10A and having the same height HT for use in a topper. However, buckling member 36tt has additives 80 mixed into the material used to make the buckling member 36tt. Regardless of the material used to make the buckling member 36tt, additives 80 such as slow-release fragrances, antimicrobial additives such as copper and silver, color additives, thermochromic materials, bed bug inhibitors, conductive additives such as graphite, aluminum powder, silicon carbide and diamond dust may be incorporated into the material. Density reducing agents such as fumed silica or gas generating materials may be added too.

[0131] If desired, buckling member 36tt with additives 80 may be used in place of any of the topper buckling members including buckling member 36f of FIGS. 10-10A.

[0132] FIG. 11 illustrates an alternative embodiment of product such as a mattress 10g having a foam core 94 and a topper 110 comprising a topper middle section 106 and two topper end sections 108. The topper 110 resides on top of the foam core 94. Each topper end section 108 comprises a piece of foam located adjacent to the topper middle section 106 and may be secured to the topper middle section 106 using adhesive or any known method. The foam pieces 108 may be glued or otherwise adhered to an upper surface 99 of the foam core 94. As shown in FIG. 11, a conventional padding or cushioning layer 14 resides above the topper 110.

[0133] The topper middle section 106 comprises a mini pocketed buckling member assembly 102 which has a bottom scrim sheet 100 which may be glued or otherwise adhered to an upper surface 99 of the foam core 94. The mini pocketed buckling member assembly 102 of topper 110 is illustrated comprising multiple strings 26g of pocketed buckling members 28g which are shorter than the strings 26 of the pocketed buckling members 28 described herein. The mini pocketed buckling member assembly 102 of topper 110 has longitudinally extending members 32 and transversely extending rows 30 of pocketed buckling members. Alternatively, the strings 26g may be offset from one another, as shown in FIG. 6.

[0134] Although FIG. 11 illustrates a topper comprising three sections, the topper may comprise any number of sections, at least one of the sections comprising strings of pocketed buckling members. For example, the topper may have two sections oriented like “his” and “hers” sections. One of the topper sections may comprise strings of pocketed buckling members and the other topper section may comprise some other material.

[0135] Alternatively, in a topper being made solely of pocketed buckling members, some of the buckling members may have one or more additives as the term is defined herein while other buckling members have fewer or no additives. The different amount of additives may result in a posturized topper having different sections or areas of different firmness due to the different composition of the buckling members of the topper.

[0136] Referring to FIG. 11, the foam core 94, topper 110 and padding or cushioning layer 14 are completely enclosed within an upholstered covering 20. As shown in FIG. 10, fully assembled, the mattress 10g has a length “L” defined as the linear distance between opposed end surfaces 22 (only one being shown in FIG. 11). Similarly, the assembled mattress 10g has a width “W” defined as the linear distance between opposed side surfaces 24 (only one being shown in FIG. 11). In the product shown in FIG. 11, the length is illustrated as being greater than the width. However, it is within the scope of the present invention that the length and width may be identical, as in a square product.

[0137] FIG. 12 illustrates an alternative embodiment of product such as a mattress 10h having a foam core 94 and a posturized topper 114 comprising different areas or regions of different firmness due to the buckling members inside the pockets of the posturized topper 114. The posturized topper 114 resides on top of the foam core 94 and may be glued or otherwise adhered to an upper surface 99 of the foam core 94. Although not shown, one or more conventional padding or cushioning layers may reside above the topper 114. The posturized topper 114 comprises two end sections 132 and a middle section 134, such a posturized topper may have any number of sections.

[0138] The mini pocketed buckling member assembly 128 of topper 114 is illustrated in FIG. 12 comprising multiple strings 26*h* of pocketed buckling members 28*h* which are shorter than the strings 26 of the pocketed buckling members 28 described herein. As seen in FIG. 12, the strings 26*h* of pocketed buckling members 28*h* may be joined so that the individually pocketed buckling members 28*h* are aligned in transversely extending rows 30 and longitudinally extending members 32. Alternatively, the strings 26*h* may be offset from one another, as shown in FIG. 6. Although the strings 26*h* are shown in FIG. 12 extending transversely or from side to side, they may be extend longitudinally.

[0139] Although FIGS. 11 and 12 illustrate a foam core 94 on which the toppers 110, 114 reside, respectively, the core may be any conventional core including a pocketed spring core, pocketed buckling member core or combination of known elements.

[0140] FIG. 13 illustrates a pocketed comfort layer 116 made of two pieces of material, a first or upper ply 122 and a second or lower ply 124 which are welded together with circular weld seams 130. The circular weld seams 130 comprise arcuate weld segments 126 with gaps 131 therebetween. Instead of mini coil springs as disclosed in U.S. Pat. Nos. 9,943,173; 9,968,202 and 10,813,462, each patent of which is fully incorporated by reference, mini buckling members 128 are located inside pockets 120 defined by the circular weld seams 130.

[0141] The mini buckling members 128, like the other buckling members described herein, may be made of any material described herein and may have any additives described herein. Similarly, mini buckling members 128, like the other buckling members described herein, may be any desired shape or configuration described herein.

[0142] FIG. 14 illustrates a pocketed comfort layer 156 made of two pieces of material, a first or upper ply 122 and a second or lower ply 124 which are welded together with intersecting linear weld seams 170. The linear weld seams 170 comprise straight weld segments 168 with gaps 177 therebetween. Instead of mini coil springs, mini buckling members 128 are located inside pockets 184 defined by the intersecting linear weld seams 170.

[0143] Each of the buckling members shown or described herein may be made by at least one of the following methods: extrusion and cut to a desired member height or injection molding.

[0144] FIG. 15 illustrates a buckling member 36*b*, as shown in FIG. 7B, filled with foam 172; a buckling member 36*c*, as shown in FIG. 7C, filled with foam 172; a buckling member 36*d*, as shown in FIG. 7D, filled with foam 172; a buckling member 36*e*, as shown in FIG. 7E, filled with foam 172 and a buckling member 36*f*, as shown in FIG. 7F, filled with foam 172. Although not shown, buckling members 36*a*, 36*g* and 36*h* may be filled with foam. Similarly, any of the other buckling members shown or described herein including those shown in FIGS. 71 through 7N, may be filled with foam. Although FIG. 15 shows the buckling members being filled with the same foam; each may be filled with any known foam. Each buckling member may be fully filled or partially filled with foam. Although each of the foam filled buckling members shown in FIG. 15 is shown being a certain height, each of them may be any desired height. The drawings are not intended to be limiting.

[0145] FIG. 16 illustrates a perspective view of the portion of a string 176 of pocketed foam filled buckling members

178 in a relaxed condition under no external load. As best illustrated in FIG. 16, each string 176 of pocketed buckling members 178 comprises a row of interconnected fabric pockets 34. Each of the fabric pockets 34 contains a single foam filled buckling member 176 and no other items. In other words, each of the fabric pockets 34 contains solely a single foam filled buckling member 176.

[0146] Preferably, one piece of fabric is used to create the pockets 34 of the string 176 of pocketed foam filled buckling members 178, the piece of fabric being folded over onto itself around the foam filled buckling members 178. As best shown in FIGS. 16 and 17, opposite sides or plies 47, 49 of the fabric are sewn, welded or otherwise secured together to create a longitudinal seam 50 and a plurality of separating or transverse seams 52. FIG. 16 illustrates ply 47 being closest to the reader and ply 49 being behind the foam filled buckling members 178.

[0147] As best shown in FIG. 16, opposed edges 56 of the piece of fabric used to create the string 176 of pocketed foam filled buckling members 178 are aligned and spaced from the longitudinal seam 50 a distance indicated by numeral 58. Although the drawings indicated the longitudinal seam 50 being below the free edges 56 of the piece of fabric, the longitudinal seam 50 may be above the free edges 56 of the piece of fabric.

[0148] As shown in FIGS. 16 and 17, the piece of fabric used to create the string 176 has a plurality of upper ears 53 and a plurality of lower ears 83. When the ears are collapsed, the string 176 has a generally planar top surface 60 in a top plane P1 and a parallel generally planar bottom surface 62 in a bottom plane P2. The linear distance between the top and bottom surfaces 60, 62 of the string 176 defines a height "H" of the string 176. This linear distance further defines the height H of the pocketed foam filled buckling member assembly 12 because each of the strings 176 has the same height. However, it is within the scope of the present invention that different strings of a pocketed foam filled buckling member assembly have different heights.

[0149] As shown in FIG. 17, in one embodiment, the transverse seams 52 of string 176 separating adjacent pockets extend from the top of an upper ear 53 of fabric to the bottom of a lower ear 83 of fabric. However, the transverse seams 52 may be any desired length.

[0150] An ultrasonically weldable fabric is used to create the string of pocketed foam filled buckling members 176. The ultrasonically weldable fabric is permeable to airflow through the fabric itself due to the nature of the fabric. Air moves between adjacent fabric pockets 38 and into and out of the string 176 through the ultrasonically weldable fabric.

[0151] As best shown in FIG. 16, the longitudinal seam 52 comprises multiple spaced linear weld segments 64 formed using an ultrasonic welding horn and anvil (not shown) as known in the art. At least some of the longitudinal seams 52 of a string may not be segmented or be only partially segmented. For example, the longitudinal seam 52 of a string may not be segmented at all.

[0152] As best shown in FIG. 17, each transverse or separating seam 52 comprises multiple spaced linear weld segments 66 formed using an ultrasonic welding horn and anvil (not shown) to join the opposed sides or plies 47, 49 of the ultrasonically weldable fabric. Again, at least some of the transverse or separating seams 52 of a string may not be segmented or may be only partially segmented. For example, one or more transverse seams 52 of a string may

be partially segmented or not be segmented at all. Although the weld segments in the embodiments shown are illustrated as being heat-welded spaced rectangular-shaped segments, any of the seam segments may be other shapes, such as spaced dots, ovals or triangles of any desired sizes.

[0153] As best shown in FIGS. 16 and 17, the foam filled buckling member 176 is shown as being cylindrically shaped member like a can. Foam filled buckling member 176 is illustrated having an outer wall 180 having an outer surface 182 and an inner surface 184, the distance between which defines the thickness “T” of the outer wall 180. The foam filled buckling member 176 has a generally planar upper surface 167 and a generally planar lower surface 169 which define the height “H” of the foam filled buckling member 176 which is approximately the same as the height of the string 176. Although the drawings illustrate one thickness “T” of outer wall 180, the outer wall 180 may be any desired thickness.

[0154] As best show in FIGS. 16 and 17, inside each of the foam filled buckling members 176 is a foam cylinder 190 having an upper surface 192 and a lower surface 194.

[0155] FIG. 17 illustrates one embodiment of pocketed foam filled buckling member assembly 186 and border strings 6 of pocketed border buckling members 200 extending around a pocketed foam filled buckling member assembly 186 of strings 176 joined together as shown in FIG. 16A. Each of the border strings 6 may extend from side-to-side or transversely and/or from end-to-end or longitudinally. Each border string 6 comprises a single piece of fabric made into a plurality of individual border pockets 82 of the same height HHH defined as the linear distance between an upper surface 84 of the border pocket 82 in plane P1 and a lower surface 86 of the border pocket 82 located in plane P2. The height HHH of the border strings 6 is preferably identical to the height H of the pockets 34 of the pocketed foam filled buckling member assembly 186. However, these heights may be different. A border buckling member 200 having a diameter D is located in each of the border pockets 82. However, the border buckling member 200 may have any desired diameter. A foam cylinder 202 is located in each of the border buckling members 200, in one embodiment. In another embodiment, a coil spring may be located in each of the border pockets 82, as shown generally in FIG. 4A.

[0156] The foam of the foam cylinders 202 of the border strings 6 is shown being a different foam than the foam of the foam cylinders 190 of the strings 176. However, the foam of the foam cylinders 202 of the border strings 6 may be the same foam as the foam of the foam cylinders 190 of the strings 176.

[0157] Referring to 16A, the strings 176 of pocketed foam filled buckling members 178 may be joined in a pocketed foam filled buckling member assembly 186 so that the individually pocketed foam filled buckling members 178 are aligned in transversely extending rows 30 and longitudinally extending members 32. Alternatively, the strings 176 may be offset from one another in a pocketed foam filled buckling member assembly, the concept of which is shown in FIG. 6.

[0158] FIG. 18 illustrates one method of making a foam filled buckling member 210. A cylinder of foam 204 is precut to size having a diameter D approximately the same distance as an inner diameter DD of an empty buckling member 206. The foam cylinder 204 is then inserted into the empty buckling member 206 as indicated by arrow 205. Although FIG. 18 shows the foam cylinder 204 being

lowered into the empty buckling member 206, the foam cylinder 204 may be raised into the empty buckling member 206. The foam cylinder 204 may be secured inside the buckling member 206 with glue or any known manner. Although the drawings show a foam cylinder 204 of a uniform height H approximately the same height as the height HH of the buckling member 206, the foam cylinder may be a different height than the height of the buckling member.

[0159] FIG. 19 illustrates another method of making a foam filled buckling member 210. This method involves pouring and/or inserting liquid foam components 212 from a container 214 into the interior 216 of an empty buckling member 218. A floor 220 prevents the liquid foam components 212 from passing through the empty buckling member 218. The liquid foam components 212 cure inside the interior 216 of the buckling member 218. Due to the curing of the liquid foam components 212, the resultant solid foam 222 may extend above a generally planar upper edge 214 of the buckling member 218. A user may cut the solid foam 222 extending above the upper edge 214 of the buckling member 218 with a cutter 224. Although one type of cutter 224 is illustrated, any known cutting instrument may be used. After the cutting has been completed, the solid foam 222 has an upper surface 226 generally coplanar with the upper edge 214 of the buckling member 218. The floor 220 may be temporary and removable or remain as part of the foam filled buckling member 210.

[0160] Although not shown, another method of introducing liquid foam components for curing inside a buckling member is disclosed in U.S. patent application Ser. No. 18/645,894, which is fully incorporated by reference herein.

[0161] FIG. 20 illustrates a mattress 240 comprising a foam layer 230 in which a plurality of buckling members 232 are incorporated. The foam layer 230 has a generally planar upper surface 234 and a generally planar lower surface 236, the linear distance between which defines the height H of the foam layer 230. The foam layer 230 is surrounded by an upholstered covering 238. Additional foam, gel or comfort layers may be incorporated into mattress 240 or any other mattress shown or described herein.

[0162] In the embodiment illustrated in FIG. 20, the buckling members 232 are arranged in longitudinally extending members 233 and transversely extending rows 235. All the buckling members 232 are identical and cylindrical having an open top and open bottom. As shown in FIG. 20, the upper edges 244 of the buckling members 232 are coplanar with the generally planar upper surface 234 of the foam layer 230. As seen in FIG. 20, the lower edges 246 of the buckling members 232 are aligned above the lower surface 236 of the foam layer 230. Although not shown, the buckling members 232 may be the full height of the foam layer 230.

[0163] Although FIG. 20 shows cylindrical buckling members 232, any of the other shapes of buckling members shown or described herein may be incorporated into a foam layer for use in any bedding or seating product including, but not limited to, mattresses. This is true for any of the mattresses showing cylindrical buckling members 232 incorporated therein.

[0164] Although the mattress 240 shows one foam layer, foam layer 230 having buckling members 232 incorporated therein may be incorporated into any mattress having any

number of layers of foam and or coil springs, including pocketed coil springs. The drawings are not intended to be limiting.

[0165] FIG. 21 illustrates a mattress 240a having a posturized foam layer 230a having a central section 248 between two end sections 250. In the central section 248 only, a plurality of buckling members 232 are incorporated. Each of the end sections 250 has no buckling members. Due to the presence of the buckling members 232, the central section 248 has a different firmness than the end sections 250 lacking buckling members. Typically, the central section 248 is firmer than the end sections 250 due to the buckling members 232 in the central section 248.

[0166] The foam layer 230a has a generally planar upper surface 234a and a generally planar lower surface 236a, the linear distance between which defines the height H of the foam layer 230a. The foam layer 230a is surrounded by an upholstered covering 238.

[0167] In the embodiment illustrated in FIG. 21, the buckling members 232 are arranged in longitudinally extending members 233 and transversely extending rows 235. The buckling members 232 are identical and cylindrical having an open top and open bottom. As shown in FIG. 21, the upper edges 244 of the buckling members 232 are coplanar with the generally planar upper surface 234a of the foam layer 230a. As seen in FIG. 21, the lower edges 246 of the buckling members 232 are aligned above the lower surface 236a of the foam layer 230a. Although FIG. 21 shows cylindrical buckling members 232, any of the other shapes of buckling members shown or described herein may be incorporated into a foam layer for use in any bedding or seating product including, but not limited to, mattresses. Although not shown, the buckling members 232 may be the full height of the foam layer 230a.

[0168] Although the mattress 240a shows one foam layer 230a having buckling members 232 incorporated therein, foam layer 230a may be incorporated into any mattress having any number of layers of foam and or coil springs, including pocketed coil springs. The drawings are not intended to be limiting.

[0169] FIG. 22 illustrates a mattress 240b having a posturized foam layer 230b having a central section 252 between two end sections 254. In the central section 252 only, a plurality of buckling members 232 are incorporated. Each of the end sections 254 has no buckling members. Due to the presence of the buckling members 232, the central section 252 has a different firmness than the end sections 254 lacking buckling members. Typically, the central section 252 is firmer than the end sections 254 due to the buckling members 232 in the central section 252.

[0170] The foam layer 230b has a generally planar upper surface 234b and a generally planar lower surface 236b, the linear distance between which defines the height H of the foam layer 230b. Although not shown, the buckling members 232 may be the full height of the foam layer 230b. The foam layer 230b is surrounded by an upholstered covering 238.

[0171] In the embodiment illustrated in FIG. 22, the buckling members 232 are arranged in a particular pattern. This pattern may be changed as desired. All the buckling members 232 are identical and cylindrical having an open top and open bottom. As shown in FIG. 22, the upper edges 244 of the buckling members 232 are coplanar with the generally planar upper surface 234b of the foam layer 230b.

As seen in FIG. 22, the lower edges 246 of the buckling members 232 are aligned above the lower surface 236b of the foam layer 230b. Although FIG. 22 shows cylindrical buckling members 232 any of the other shapes of buckling members shown or described herein may be incorporated into a foam layer for use in any bedding or seating product including, but not limited to, mattresses.

[0172] Although the mattress 240b shows one foam layer, foam layer 230b having buckling members 232 incorporated therein may be incorporated into any mattress having any number of layers of foam and or coil springs, including pocketed coil springs. The drawings are not intended to be limiting.

[0173] FIGS. 23 and 24 illustrate a mattress 240c having a posturized foam layer 230c having a head section 256, a middle section 258 and a foot section 260 of different firmness due to different buckling members incorporated into the different sections. In the head section 256 only, a plurality of buckling members 262 are incorporated. In the middle section 258 only, a plurality of buckling members 264 are incorporated. In the foot section 260 only, a plurality of buckling members 266 are incorporated. As best shown in FIG. 24, the buckling members 262, 264 and 266 are all the same height. As best shown in FIG. 24, the thickness of the outer walls of the buckling members 262 within the head section 256 is greater than the thickness of the outer walls of the buckling members 266 of the foot section 260 so the head section 256 is firmer than the foot section 260. Similarly, the thickness of the outer walls of the buckling members 266 within the foot section 260 is greater than the thickness of the outer walls of the buckling members 264 of the middle section 258 so the foot section 260 is firmer than the middle section 258. In general, the thickness of the outer walls of the buckling members vary to impart different firmness to the different sections.

[0174] The foam layer 230c has a generally planar upper surface 234c and a generally planar lower surface 236c, the linear distance between which defines the height H of the foam layer 230c. Although not shown, the buckling members 232 may be the full height of the foam layer 230c. As shown in FIG. 23, the foam layer 230c is surrounded by an upholstered covering 238.

[0175] In the embodiment illustrated in FIGS. 23 and 24, the buckling members 262, 264, 266 are arranged in longitudinally extending members 233 and transversely extending rows 235. all the buckling members 262, 264, 266 are cylindrical having an open top and open bottom. As shown in FIG. 24, the upper edges 244 of the buckling members 262, 264, 266 are coplanar with the generally planar upper surface 234c of the foam layer 230c. As seen in FIG. 24, the lower edges 246 of the buckling members 262, 264, 266 are aligned above the lower surface 236c of the foam layer 230c.

[0176] Although FIGS. 23 and 24 show cylindrical buckling members 262, 264, 266, any of the other shapes of buckling members shown or described herein may be incorporated into a foam layer for use in any bedding or seating product including, but not limited to, mattresses.

[0177] Although the mattress 240c shows one foam layer, a foam layer 230c having buckling members 262, 264, 266 incorporated therein may be incorporated into any mattress having any number of layers of foam and or coil springs, including pocketed coil springs. Although the foam layer 230c shows buckling members 262, 264, 266 having outer walls of different thickness incorporated therein; the draw-

ings are not intended to be limiting. For example, the thickness of the outer walls of the buckling members of the head and foot sections may be less than the thickness of the outer walls of the buckling members of the middle section. In general, the thicker the outer walls of the buckling members within a section, the greater firmness the section will have due to the buckling members therein.

[0178] Although each of the mattresses **240**, **240a**, **240b**, **240c** and **240d** is shown being a certain height, the drawings are not intended to be limiting. Any one of these mattresses comprising a foam layer embedded with buckling members may be of any desired height and incorporated into a known bedding or seating product. Similarly, although the drawings show each of the buckling members being shorter than the height of the foam layer, the buckling members may be the same height as the height of the foam layer into which they are incorporated.

[0179] FIG. 25 illustrates a mattress **240d** comprising a pocketed spring assembly **270** including a base **272**. The mattress **240** further comprises a foam layer **274** in which a plurality of buckling members **276** are incorporated. The foam layer **274** has a generally planar upper surface **278** and a generally planar lower surface **280**, the linear distance between which defines the height H1 of the foam layer **274**. The foam layer **274** and pocketed spring assembly **270** is surrounded by an upholstered covering **238**. Upper edges **282** of the buckling members **276** are co-planar with the upper surface **278** of foam layer **274**. Although each of the buckling members **276** are illustrated being shorter than the height of the foam layer **274**, they may be any desired height including the full height of the foam layer.

[0180] The various embodiments of the invention shown and described are merely for illustrative purposes only, as the drawings and the description are not intended to restrict or limit in any way the scope of the claims. Those skilled in the art will appreciate various changes, modifications, and improvements which can be made to the invention without departing from the spirit or scope thereof. The invention in its broader aspects is therefore not limited to the specific details and representative apparatus and methods shown and described. Departures may therefore be made from such details without departing from the spirit or scope of the general inventive concept. The invention resides in each individual feature described herein, alone, and in all combinations of any and all of those features. Accordingly, the scope of the invention shall be limited only by the following claims and their equivalents.

What is claimed is:

1. A bedding or seating product comprising:

a pocketed buckling member assembly comprising a plurality of parallel strings of pocketed buckling members, each of said strings being joined to at least one adjacent string, each of said strings comprising a piece of fabric surrounding a plurality of buckling members, first and second opposed plies of fabric being on opposite sides of the buckling members, a plurality of pockets formed along said string by a longitudinal seam and transverse seams joining said first and second plies, a single buckling member being in each pocket, wherein each of the buckling members is a unitary member having an outer wall, an open top and an open bottom, each of the buckling members being made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene

propylene diene monomer (EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythioethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof, wherein at least some of the buckling members are at least partially filled with foam.

2. The product of claim 1 further comprising: a covering encasing said pocketed buckling member assembly.

3. The product of claim 1, wherein at least some of the buckling members contain at least one of the following additives: phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes, density reducing agents or gas generating materials.

4. The product of claim 1, wherein at least some of the buckling members have a different firmness than other buckling members.

5. The product of claim 4, wherein the pocketed buckling member assembly has a uniform height.

6. The product of claim 1, wherein at least some of the buckling members contain conductive additive solids, powders, particles or flakes.

7. The product of claim 1, wherein at least some of the buckling members are fully filled with foam.

8. A pocketed buckling member assembly for a bedding or seating product, said pocketed buckling member assembly comprising:

a plurality of parallel strings of pocketed buckling members, each of the strings being joined to at least one adjacent string, each of said strings comprising a plurality of interconnected pockets, each of the pockets containing at least one buckling member encased in fabric, the buckling member filling the pocket, the fabric being connected to itself along a longitudinal seam and having first and second opposed plies of fabric on opposite sides of the buckling members, a plurality of pockets formed along said string by transverse seams joining said first and second plies,

wherein each of the buckling members is a unitary member being made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythioethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof and

wherein at least some of the buckling members are filled with foam.

9. The pocketed buckling member assembly of claim 8, wherein at least some of the buckling members contain at least one of the following additives: phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive addi-

tive solids, powders, particles, flakes, density reducing agents or gas generating materials.

10. The pocketed buckling member assembly of claim **9**, wherein at least some of the buckling members contain phase change materials.

11. The pocketed buckling member assembly of claim **8**, wherein at least some of the buckling members contain conductive additive solids, powders, particles or flakes.

12. The pocketed buckling member assembly of claim **8**, wherein at least some of the buckling members have an outer wall, an open top and an open bottom.

13. The pocketed buckling member assembly of claim **8**, wherein the pocketed buckling member assembly has a uniform height and each buckling member in each pocket has the same height.

14. The pocketed buckling member assembly of claim **8**, wherein at least some of the buckling members have an outer wall which is circular in cross-section.

15. A pocketed buckling member assembly for a bedding or seating product, said pocketed buckling member assembly comprising:

a plurality of parallel strings of pocketed buckling members, each of the strings being joined to an adjacent string, each of the strings comprising a plurality of interconnected pockets made from fabric, each of the pockets being approximately the same height, each pocket containing a single buckling member having the same height as the pocket, the piece of fabric being joined to itself along a longitudinal seam and having first and second opposed plies of fabric on opposite sides of the buckling members, the fabric of said first and second plies being joined by transverse seams; some of the strings having buckling members made of a different material than the buckling members of other strings, and

wherein each of the buckling members is a unitary member having hollow passages and being made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythioethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene and polypropylene and wherein each buckling member is at least partially filled with foam.

16. The pocketed buckling member assembly of claim **15**, wherein at least some of the buckling members contain at least one of the following additives: phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes, density reducing agents or gas generating materials.

17. The pocketed buckling member assembly of claim **15**, wherein at least some of the buckling members have an outer wall, an open top and an open bottom defining a passage.

18. The pocketed buckling member assembly of claim **15**, wherein the buckling members are the same height.

19. The pocketed buckling member assembly of claim **15**, wherein at least some of the buckling members contain conductive additive solids, powders, particles or flakes.

20. The pocketed buckling member assembly of claim **15**, wherein at least some of the buckling members contain phase change materials.

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